MIL-R-83403(USAF)
1 May 1973

MILITARY SPECIFICATION
ROCKET MOTOR SR110-AD-1

1. SCOPE

1.1 Scope. This specification covers the requirements for the Rocket Motor SR110-AD-1.

1.2 Classification. The Rocket Motor SR110-AD-1 is an internal burning, solid-propellant rocket motor which is approximately four-inches in diameter and 79-inches long. As the propulsion unit for Meteorological Probes PWN-10A, PWN-10B, PWN-11A or the PWN-12A, this rocket motor has been designed to boost small diameter non-propulsive darts to altitudes of approximately 240,000-feet and 370,000-feet. This rocket motor contains a separable igniter, four fins, and a forward end mating structure for the Ogive, Telemetric Instrumentation Groups MXU-621/A, MXU-622/A, MXU-623/A and MXU-642/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of exact issue shown, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-23702	10 May 63	Polysulfide, Polymer, Liquid for Rocket Motor, 3.0 Inch, Mark 32 Mod 0
MIL-Q-23703	24 Feb 65	Quinonedioxime, for Rocket Motor, 3.0 Inch, Mark 32 Mod 0
MIL-S-23704	26 Oct 64	Sulfur, Flowers of, for Rocket Motor, 3.0 Inch, Mark 32 Mod 0
MIL-D-23705	24 Feb 65	Diphenylguanldine, Symmetrical (DPG) Rocket Motor, 3.0 Inch, Mark 32 Mod 0

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MIL-R-25533	26 Jun 59	Rocket Motor, Aeronautical, Model Specification for (Outline and Instructions for Preparation)
MIL-R-25534	26 Jun 59	Rocket Motor, Aeronautical, Qualification Test for
MIL-R-25535	26 Jun 59	Rocket Motor, Aeronautical, Preliminary Flight Rating Test for
MIL-R-25536	26 Jun 59	Rocket Motor, Aeronautical, Acceptance Test for
MIL-R-83404(USAF)	1 May 73	Rocket Motor Manufacturing, Rocket Motor SR110-AD-1
MIL-P-83405(USAF)	1 May 73	Propellant, Composite ADC-109
MIL-L-83066	1 Jun 68	Liner and Inhibitor Material for Rocket Motors

STANDARDS

Military

MIL-STD-105	20 Mar 66	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	28 Dec 64	Marking for Shipment & Storage
MIL-STD-746	31 Jul 63	Radiographic Testing Requirements for Cast Explosives
MIL-STD-1167	1 Feb 68	Ammunition Data Card
MIL-STD-1168	30 Jun 65	Lot Numbering of Ammunition

HANDBOOKS

Military

MIL-HDBK-53	30 Jun 65	Guide for Sampling Inspection

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DRAWINGS

Air Force

70J2222	Rocket Motor SR110-AD-1
71D48862	Rocket Motor Packing
71D48868	Igniter Assembly

PUBLICATIONS

Bureau of Naval Weapons

OS 9410B	Aluminum Powder
OS 11327	Ammonium Perchlorate
OS 11645	Magnesium Oxide

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Qualification and acceptance. The qualification and acceptance of the rocket motor shall be in accordance with the tests specified in Section 4 of this specification.

3.2 Mock up. Not applicable.

3.3 Installation changes. Not applicable.

3.4 Performance characteristics. All performance characteristics are based upon the use of operational nozzles.

3.4.1 Rocket motor operating regimes. The rocket motor ignition and operating envelope shall be as shown in Figure 1.

3.4.1.1 Airspeed. Not applicable.

3.4.1.2 Altitude. The rocket motor shall ignite and operate satisfactorily at altitudes up to 12,000 feet.

3.4.1.3 Temperature. The rocket motor shall ignite and operate satisfactorily under the following conditions.

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3.4.1.3.1 Stabilized condition. The rocket motor shall perform satisfactorily after static exposure at a maximum temperature of +140°F for a period long enough to condition the entire mass of the motor to +140°F, or after exposure at a minimum temperature of -40°F for a period long enough to condition the entire mass of the motor to -40°F.

3.4.1.3.2 Gradient condition.

a. The rocket motor shall perform satisfactorily when it has been conditioned to a minimum ambient temperature of +140°F and then fired after exposure to a maximum ambient temperature of -40°F when the maximum temperature gradient exists within the grain; and when it has been conditioned to a maximum ambient temperature of -40°F and then fired after exposure to a minimum ambient temperature of +140°F when the maximum temperature gradient exists within the grain.

b. The rocket motor shall perform within the envelope of performance defined by Table I between the temperature limits of -40°F after it has been exposed to extreme temperatures for periods of time as shown in Figure 2.

3.4.1.3.3 Flight operation. The rocket motor shall perform satisfactorily at altitudes of 0 to 12,000 ft and at velocities of 0 to 6,000 ft/sec. It is estimated that the motor will operate satisfactorily at any altitude. The maximum velocity limits have not been determined.

3.4.1.4 Attitudes. The rocket motor shall perform satisfactorily in any attitude.

3.4.2 Ratings.

3.4.2.1 Sea level. The performance ratings at standard sea level static conditions shall be as listed in Table I.

3.4.3 Performance curves. Performance shown in Figures 3 through 16 inclusive constitute part of this specification. These curves indicate performance attained under specified conditions. Dotted curves are super-imposed on each curve applicable, showing the tolerance on performance.

3.4.3.1 Typical sea level chamber pressure and thrust versus time curves are shown on the following figures:

Figure 3	-40°F
Figure 4	59°F
Figure 5	140°F.

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Table I. Performance Ratings at Standard Sea Level Static Conditions

	Conditioned Temp °F	Nominal Rating	Limits Magnitude Max	Limits Magnitude Min	Percent* Max	Percent* Min
Average Chamber Pressure (psia)	-40	1082.3	1221.4	943.2	112.9	87.1
	+59	1211.6	1350.7	1072.5	111.5	88.5
	+140	1313.0	1452.1	1173.9	110.6	89.4
Average Thrust (lb)	-40	3640.2	4028.4	3252.0	110.7	89.3
	+59	4020.9	4409.1	3632.7	109.7	90.3
	+140	4406.9	4795.1	4018.7	108.8	91.2
Action Time Total Impulse (lb-sec)	-40	8446.5	9039.0	7854.0	107.0	93.0
	+59	8511.2	9103.7	7918.7	107.0	93.0
	+140	8588.5	9181.0	7996.0	106.9	93.1
Action Time (sec)	-40	2.32	2.44	2.20	105.2	94.8
	+59	2.11	2.23	1.99	105.7	94.3
	+140	1.94	2.06	1.82	106.2	93.8
Ignition Delay (sec)	-40	0.0079	0.0157	0.	198.7	0.
	+59	0.0077	0.0155	0.	201.3	0.
	+140	0.0099	0.0177	0.0021	177.0	21.2
Maximum Thrust (lb)	-40	5179.6	5969.5	4389.7	115.3	84.7
	+59	5523.6	6313.5	4733.7	114.3	85.7
	+140	6013.2	6803.1	5223.3	113.1	86.9
Maximum Chamber Pressure (psia)	-40	1432.8	1593.7	1271.7	111.2	88.8
	+59	1542.9	1703.8	1382.0	110.4	89.6
	+140	1643.6	1804.5	1482.7	109.8	90.2

*Based on nominal rating

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3.4.3.2 Curves showing the variation in rocket motor performance with temperature at sea level are shown on the following figures:

Figure 6	Average Thrust
Figure 7	Action Time Total Impulse
Figure 8	Action Time
Figure 9	Average Chamber Pressure
Figure 10	Ignition Delay.

3.4.3.3 Curves showing the variation in rocket motor performance with action time at sea level are shown in the following figures:

Figure 11	Average Thrust
Figure 12	Action Time Total Impulse
Figure 13	Average Chamber Pressure.

3.4.3.4 Typical chamber pressure and thrust transients during ignition at sea level are shown on the following figures:

Figure 14	-40°F
Figure 15	59°F
Figure 16	140°F.

3.4.4 Performance variation.

3.4.4.1 Thrust. The greatest value of maximum ignition thrust at sea level is 4,700 pounds at +140°F and the least value of maximum ignition thrust is 3,255 pounds at -40°F. The greatest value of maximum thrust is 6,803 pounds in the temperature range from -40°F to 140°F and occurs at 140°F.

3.4.4.2 Pressure. The greatest value of maximum ignition pressure is 1780 psia at +140°F and the least value of maximum pressure is 1272 psia at -40°F. The greatest value of maximum chamber pressure is 1805 psia in the temperature range from -40°F to 140°F and occurs at 140°F.

3.5 Environmental and load factors.

3.5.1 Environmental conditions. The rocket motor and packaged igniter shall not suffer any detrimental effects during or after exposure to the following conditions.

3.5.1.1 Storage temperature range and attitude. The rocket motor and packaged igniter, under field storage conditions, shall not suffer any detrimental effects when exposed to the temperature range of -40°F to +140°F and when stored in any attitude.

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3.5.1.2 Limiting exposure time. The limiting exposure time shall not be less than 45-days at either a maximum temperature of -40°F or a minimum temperature of +140°F.

3.5.1.3 Life. The maximum storage life of the rocket motor at 70°F +30°F shall be 3-years.

3.5.1.4 Vibration. The rocket motor shall perform within the limits specified in Table I after being subjected to vibration test as specified in MIL-R-25534. In addition, the rocket motor shall operate within specifications, and shall flight test successfully after transportation by common carrier for distances in excess of 250-miles.

3.6 The qualification test method shall be per MIL-R-25534.

3.6.1 Explosive classification. The contractor shall request I.C.C. shipping classification.

3.7 Weight of the rocket motor. Prior to firing, the weight of the rocket motor shall not be less than 49.0 pounds nor be more than 52.2 pounds. The nominal weight of specific items is as follows:

Component	Nominal Weight, Pounds
Motor Case (with fins, nozzle, and nozzle retainer)	10.63
Head Cap	1.37
Propellant	37.51
Liner (boots and inhibitor)	.69
Sub-total:	50.20
Igniter	.40
Total:	50.60.

3.8 Components.

3.8.1 Grain. The grain shall be free of physical defects.

3.8.1.1 Propellant. The propellant shall be composed of the materials shown in Table II, each of which meets the requirements of the applicable publication listed herein. The nominal ratio of the components is also shown in Table II. However, the supplier may vary the ratios, as necessary, to achieve the performance requirements of this specification. The particle size of the Ammonium Perchlorate shall be as shown in Table III. The strain at maximum stress of cured propellant shall be 80-pounds per square inch (psi) or more when tested in accordance with Section 4 of MIL-P-83405(USAF). The propellant shall be a product free from

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Table II. Propellant Composition

Material	Purpose	Publication	Parts by Weight (nominal)
Polysulfide Polymer, Liquid	Fuel and Binder	MIL-P-27302	16.40
Quinonedioximine	Curing Agent	MIL-Q-23703	1.20
Sulfur, Flowers of	Curing Catalyst	MIL-S-23704	0.10
Diphenylguanidine	Curing Accelerator	MIL-D-23705	0.10
Ammonium Perchlorate* (as received)	Oxidizer	OS 11327	46.20
Ammonium Perchlorate* (after grinding)	Oxidizer	OS 11327	30.80
Dibutyl Phthalate	Plasticizer	----------	2.80
Magnesium Oxide	Curing Catalyst	OS 11645	0.60
Aluminum	Resonance Suppressor	OS 9410B	1.80

*See Table III.

Table III. Grain Size Distribution for Ammonium Perchlorate Blend

Characteristic	Requirements Min	Max
As received AP retained on:		
USS 18%	-0-	-0-
USS 50%	3%	11%
USS 100%	50	82
USS 140%	85	98
USS 325%	98	100
Ground AP% below:		
10 Microns	12	32
20 Microns	40	60
30 Microns	58	78

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foreign materials. It shall be uniform in quality, and manufactured in accordance with standard manufacturing procedures of the industry. As determined by X ray analysis, void limitations for the propellant grain shall be as indicated in Table IV.

3.8.1.1.1 Ballistic properties. Ballistic properties are as follows:

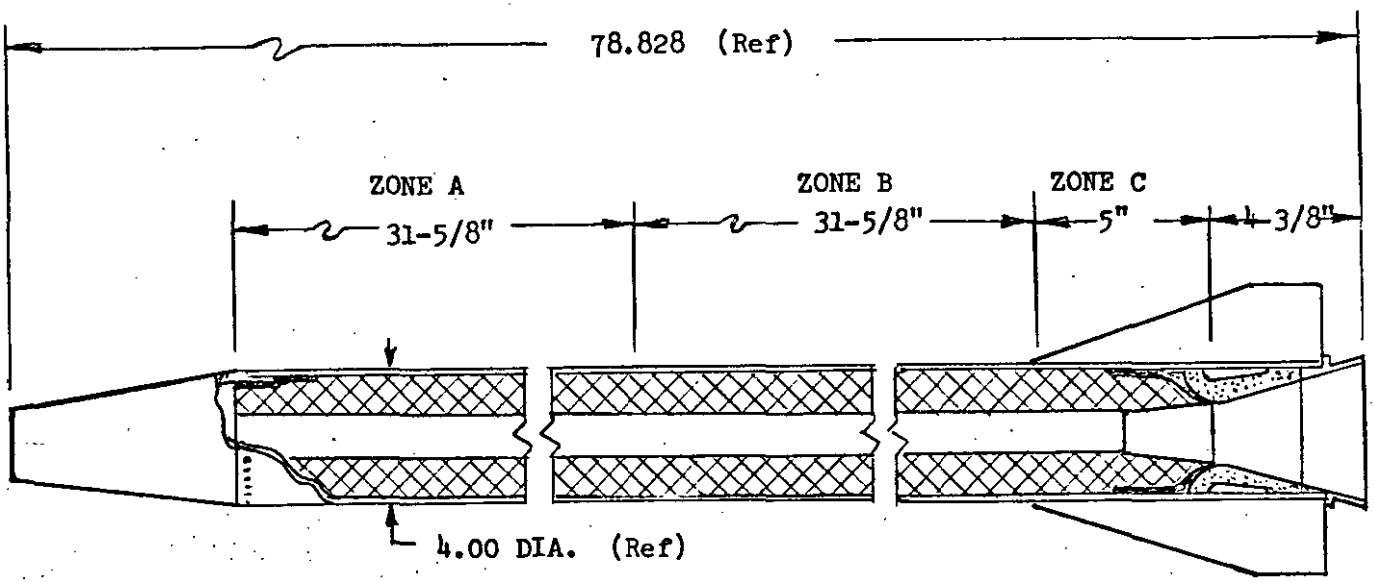
a. Constants and coefficients.

Temperature coefficient of chamber pressure	0.308 per deg Kelvin
Temperature coefficient of burn rate	0.0488 per deg Kelvin
Pressure exponent	0.435
Characteristic exhaust velocity	4702 fps
Ratio of specific heats	1.217
Heat of explosion	1200 cal/g
Adiabatic flame temperature	2784°K
Specific impulse @ 1000 psi/SL	232.11 sec

b. Equilibrium exhaust composition.

Species	Moles Gas	Mole Percent
CO	0.549396	14.5822
CO_2	0.291555	7.7315 6
H_2	0.383396	10.1669
H_2O	1.374562	36.4503
N_2	0.337047	8.9379
HCL	0.640386 9	16.9818

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Zone	Voids Inches Radial	Voids Max. Long.	Accumulated Voids* Inches Radial	Accumulated Voids* Max. Long.
A	3/8	3/8	3/8	5/8
B	3/8	3/8	3/8	1/2
C	1/4	1/4	1/4	1/4

* Accumulated voids are found by adding the radial and longitudinal dimensions of each void found in a given zone.

TABLE IV VOID LIMITATIONS

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Equilibrium exhaust composition - contd -

Species	Moles Gas	Mole Percent
MGO	0.014880	Solid
AL_2O_3	0.032876	Solid
H	0.011409	0.3025
CL	0.013053	0.3461
OH	0.011233	0.2979
O_2	0.000305	0.0081
O	0.000203	0.0055
$ALCL_2$	0.000550	0.0145
ALCL	0.000063	0.0017
ALCLO	0.000234	00.0062
$ALCL_3$	0.000145	0.0038
ALO	0.000000	0.0000
H_2S	0.015769	0.4182
COS	0.000935	0.0248
SO_2	0.093665	2.4838
SO	0.027514	0.7296
HS	0.013894	0.3684
S_2	0.005145	0.1364
S	0.000035	0.0009.

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c. A curve showing burning rate and area ratio vs chamber pressure is shown on Figure 17. The formula for deriving these curves is:

$$Kn = 2.675P_c^{0.648} \quad \text{and} \quad r = .0404P_c^{0.352}.$$

d. A curve showing variation of burn rate with rocket temperature at the initial area ratio of the rocket motor is shown on Figure 18.

3.8.1.1.2 Physical and mechanical properties. Physical and mechanical properties of the propellant are as follows:

a.	Density	0.062 lb/in^3
b.	Hardness	70 - 90 Shore A
c.	Tensile Strength	200 - 250 psi Maximum
d.	Elongation	18 - 45% Maximum
e.	Modulus	2100 - 2800 psi.

3.8.1.1.3 Autoignition temperature. The highest temperature at which a sample of cured propellant will not autoignite after one-hour, is 275°F. The highest temperature at which a sample of cured propellant will not autoignite after eight-hours is 225°F.

3.8.1.2 Restrictor and bonding agent.(Insulation).

a. The bonding agent for the forward insulation boot shall be Eastman 910. The bonding agent for the aft insulation boot shall be .Stycast 2651.

b. The Liner bonding material, for the forward closure, shall conform to MIL-L-83066(USAF).

3.8.1.3 Processing. The processing of propellant shall be in accordance with MIL-R-83404(USAF).

3.8.1.4 Corrosive and toxic effects. With the exception of the items noted in 3.8.1.1.1 b., the exhaust gases are both corrosive and toxic.

3.8.2 Liner formulation and processing. The formulation and processing of the liner shall be in accordance with MIL-L-83066(USAF).

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3.8.3 Connectors and cable. It shall be possible to connect or disconnect electrical connections and to flex electrical conductors, without damage, throughout the temperature range of -40°F to +140°F.

3.8.4 Case.

3.8.4.1 Proof, yield and burst pressures. The case proof pressure at 70°F ±20°F is 1500 psi. The minimum yield pressure at +140°F is 2320 psi. The minimum burst pressure at +140°F is 2700 psi.

3.8.4.2 Maximum operating case temperature. The maximum operating case temperature attained before burnout is 200°F.

3.8.5 Nozzle. Throat area is 2.326 ±.003 square inches. The maximum allowable increase in throat area is 0.10 square inch. The nozzle expansion ratio is 5.04.

3.8.5.1 Nozzle closure blowout pressure. The blowout pressure of the nozzle closure shall be 50 psia ±15 psia.

3.8.6 Igniter. The rocket motor igniter shall be an electrically actuated metallic particle type igniter, and shall perform with a minimum of shock. It shall consist of a consumable case containing ignition material, and two electrically actuated 1 Watt/1 Ampere no-fire squibs arranged in a parallel circuit. The igniter shall conform to the drawing 71D48868.

a. The ignition material shall conform to the following formulation and material specifications:

Material	Function	Nom Parts by Wt
Cupric Oxide	Oxidizer	80
Aluminum Powder	Fuel	20.

3.8.6.1 Pre-firing check. The igniter shall incorporate a connector for continuity checks of the igniter system as necessary. The pre-firing check shall be made by connecting the igniter leads to an Alinco Continuity Tester, Model 101-5BF or equivalent. Ignition shall not occur when test current of 0.02 amperes is applied for any length of time. The igniter resistance shall be 0.50 to 0.70 ohms.

a. Safety. The igniter leads shall terminate at a self-shorting plug to prevent ignition from induced currents.

b. Igniter assembly. The igniter shall be packaged separately as specified in 71D48862 and shall be capable of insertion in the rocket motor after the motor is installed in the launch tube.

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c. Arming. The rocket motor shall be prepared for firing by inserting the igniter assembly through the nozzle as indicated on the drawings. The plug is inserted into the mating connector in the launcher firing line. The rocket may be disarmed at any time prior to firing by removing the plug from the firing line, then removing the igniter from the motor.

3.8.6.2 Autoignition. The highest temperature at which a sample of the igniter composition will not autoignite after one and one-half hours is 1000°F. The highest temperature at which a sample of the igniter composition will not autoignite after eight-hours is 420°F.

3.8.7 Thrust angle control device. Not Applicable.

3.8.8 Thrust magnitude control device. Not Applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the Rocket Motor SR110-AD-1 and Meteorological Probes PWN-10A, -10B, -11A and -12A, shall be classified as follows:

a. Qualification

(1) Rocket Motor

(2) Probe Flight Tests.

b. Lot Acceptance

(1) Rocket Motor

(2) Probe Flight Tests.

4.3 Rocket motor qualification test reports. The Contractor shall prepare a Qualification Test Report in accordance with MIL-R-25534.

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4.4 Qualification testing. Qualification testing shall be required if the vendor has not been previously qualified to manufacture the rocket motor and/or probe(s), or has not produced the rocket motor and/or probe(s) for a period of at least two (2) years.

4.4.1 Rocket motor. Rocket Motor SR110-AD-1 qualification testing shall consist of the following:

a. Twenty-seven (27) rocket motors shall be tested in accordance with Groups I through VI inclusive, of MIL-R-25534, except the temperature limits shall be -40°F to +140°F.

b. Igniter assembly tests shall be conducted to verify the requirements of paragraph 3.8.6.

4.4.2 Probe flight tests. Flight tests shall be conducted by the Government at a Government range.

4.4.2.1 Sampling plan. The sampling plan to be applied for acceptance or rejection of the flight tests shall be in accordance with MIL-STD-105, Table III A, double sampling, sample size code letter D, AQL-10%. Ten (10) of each type of probe being qualified shall be submitted for test. If any probe fails the sampling plan, quantities in the amount of ten (10) additional probes of that type shall be resubmitted for flight test until the sample has passed. Any probe previously submitted, but not flight tested, may be reworked and resubmitted as part of the subsequent ten (10) probes for flight test.

4.4.3 Accept-reject criteria.

4.4.3.1 Rocket motor. Rocket motor performance shall meet the parameters specified in Table I.

4.4.3.2 Igniter assembly. The igniter assembly shall meet the requirements of paragraph 3.8.6.

4.4.3.3 Flight tests. The performance of the probe being tested shall meet the criteria specified in paragraph 4.6.

4.5 Lot acceptance testing. Lot acceptance testing shall consist of both static tests and flight tests. The probes delivered for flight test shall be considered as part of the deliverable lot (4.7.1).

4.5.1 Rocket motor.

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4.5.1.1 Radiographic testing. Randomly select thirty-two (32) rocket motors from each lot produced. Radiographically inspect each motor in one plane in accordance with MIL-STD-746, Inspection Level 2-4T. Accept lot on one or less failures. Reject on two or more failures. In the event a lot is rejected, the remaining motors shall be 100% inspected and accepted if no defects are found. Accept-reject criteria shall be in accordance with Table IV of this specification.

4.5.1.2 Static motor firing.

4.5.1.2.1 One extra rocket motor per sampling lot (4.7.2) will be fabricated to cover the static testing specified below. The testing will be accomplished as follows:

a. One motor shall be selected at random from each sampling lot produced, radiographically inspected and static fired by the Contractor. Test temperatures of -40°F and +140°F shall apply. The test temperature shall be alternated for each lot tested for acceptance. This motor will not be considered as a part of the total deliverable quantity of a lot.

b. The motor performance must meet the parameters specified in Table I. Failure of the rocket motor constitutes rejection of the lot motors. A retest will be permissible in the event the motor ballistic data are lost because of test support equipment failure. Rework of a lot is permissible, at the Contractor's option. A reworked lot will be considered a new lot for acceptance.

4.5.2 Flight tests. Flight testing shall be conducted by the Government at a Government range.

4.5.2.1 Selection of flight test samples. Flight test sample serial numbers shall be selected by the Contracting Officer using the Table of Random Numbers, pages 21 and 22, of MIL-HDBK-53.

4.5.2.2 Sampling plan. Sampling of each lot shall be in accordance with MIL-STD-105, Inspection Level I, AQL-15%, Double Sampling Plan, Normal Inspection. A successful flight shall be as defined in paragraph 4.6 for the applicable probe.

4.6 Accept-reject criteria for flight tests. Rawin Set AN/GMD-1 or -4, and Radar Set AN/FPS-16, or equivalent, shall be used for tracking as applicable for the system being tested. Each type of system flight tested shall meet the criteria specified for that type in the paragraphs below. Based on these criteria, the Government shall

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make the determination as to the acceptability of any test. Termination of a flight test is defined as 80,000 feet above mean sea level for the PWN-10A, PWN-10B, and PWN-11A, and 50 kilometers (km) for the PWN-12A.

4.6.1 Meteorological Probe PWN-10A. Each of the following criteria must be met in order for a PWN-10A flight test to be considered successful:

a. A radar track showing payload deployment at a minimum apogee altitude of 204,000 feet above mean sea level, when launched from sea level at a QE of 80-degrees. If conditions necessitate the launching at other than 80°, the apogee altitude will be normalized to that altitude which would have been obtained at QE of 80°. The test flight will not be conducted if conditions warrant a QE of less than 78°. The altitude normalizing curve is shown in Figure 20.

b. The ballistic coefficient of the retardation device shall not exceed 0.025 pounds per square foot.

c. A usable record of temperature data shall be received at the Rawin Set AN/GMD-4 and recorded on the AN/TMQ-5 from ejection altitude to termination of flight (4.6).

d. Ranging data shall be obtained as follows:

(1) Continuous fine ranging from ejection to termination of flight.

(2) At least one verification of coarse ranging on the AN/GMD-4 at some interval between ejection and termination of flight.

4.6.2 Meteorological Probe PWN-10B. The requirements specified in 4.6.1 shall apply as criteria for the PWN-10B except subparagraph d, "Ranging Data" shall not apply.

4.6.3 Meteorological Probe PWN-11A. The requirements specified in 4.6.1 shall apply as the criteria for the PWN-11A except subparagraph c. "Ranging data" shall not apply, and subparagraph b, "the ballistic coefficient" shall be 0.040 pounds per square foot.

4.6.4 Meteorological Probe PWN-12A. The following requirements must be met in order for a PWN-12A flight test to be considered a success:

a. The inflatable sphere shall attain a minimum altitude of 320,000 feet above mean sea level when launched from sea level at a QE of 80°. If conditions necessitate the launching at other than

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80° (QE), the apogee altitude will be normalized to the altitude which would have been obtained at a QE of 80°. The test will not be conducted when conditions require a QE of less than 78°. The altitude normalizing curve is shown in Figure 21.

b. The sphere must inflate, and usable data provided from an AN/FPS-16, or equivalent, radar down to at least 50 km. Successful inflation and usability of data shall be determined by reduction of radar data using the May 1970 AFCRL ROBIN data reduction program to obtain atmospheric density and winds.

4.6.5 No-test conditions. A no-test shall be considered as a condition which indicates that the test results are not representative of the performance of the test article. Examples of conditions resulting in no-test determination are:

a. No radar track or permanent loss of radar track due to causes not associated with the system being tested.

b. Malfunction of either radar or ground telemetry (AN/GMD-4).

c. Late acquisition so that performance of the separation device cannot be determined.

4.7 Lot definition.

4.7.1 Deliverable lot. A deliverable lot shall be considered as ninety (90) probes consisting of an equal number of Rocket Motors SR110-AD-1, and Ogive, Telemetric Instrumentation Group of the type on order.

4.7.2 Sampling lot. A sampling lot shall be considered as ninety-one (91) Rocket Motor SR110-AD-1 and ninety (90) Ogive, Telemetric Instrumentation Group of the type on order.

4.7.3 Lot numbering. Lot numbering shall be in accordance with MIL-STD-1168. Each motor, ogive, instrument, and retardation device shall have serial numbers assigned. These serial numbers shall not duplicate numbers previously used.

4.7.4 Ammunition data cards. Ammunition data cards, as required in MIL-STD-1167 shall be forwarded to OOAMA (MMNTAC), Hill AFB UT, 84406. The requirement for ammunition data cards is for one card in duplicate, for each lot of motors, and one card in duplicate for each lot of dart assemblies. The lot number is to appear at the top center of the data cards. The motor serial numbers and dart assembly serial numbers are to be given in the remarks block on the back of the data cards.

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5. PREPARATION FOR DELIVERY

5.1 Method. Unless otherwise specified by the procuring activity, preparation for delivery shall be as follows:

a. The Meteorological Probes shall be packaged in accordance with the appropriate T.P.O.

b. Marking of interior and exterior containers shall be in accordance with MIL-STD-129, and the requirements of I.C.C. Regulation.

c. The Contractor shall be responsible for obtaining I.C.C. approval of his method of delivery of the rocket motor, and shall submit evidence of such approval to the procuring activity.

6. NOTES

6.1 Intended use. The rocket motor covered by this specification is intended for use with the Meteorological Probe PWN-10A, PWN-10B, PWN-11A, and PWN-12A.

6.2 Symbols and definitions. The symbols and applicable definitions used herein and in MIL-R-25533, MIL-R-25534, MIL-R-25535, and MIL-R-25536, are as specified below.

6.2.1 Definitions. Definitions of ballistic parameters are illustrated on Figure 19.

6.2.1.1 Procuring activity. The procuring activity is the Department of Defense activity which is responsible for the rocket motor contract.

6.2.1.2 Vehicle. A vehicle is an aeronautical transportation device.

6.2.1.3 Rating. A rating is the value of some characteristic of performance as specified in the model specification.

6.2.1.4 Estimate. An estimate is a predicted value of performance.

6.2.1.5 Standard conditions. Standard conditions are the values of air, temperature and pressure given in the ARDC Model Atmosphere, 1956.

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6.2.1.6 Rocket motor. A rocket motor is defined as that assembly of components which meets all requirements of this specification as modified by the individual rocket motor model specification.

6.2.1.7 JATO Unit. JATO Unit is a rocket motor used to assist in the take-off or piloted aircraft.

6.2.1.8 Magazine. A Magazine is a structure designed for the storage of explosives or ammunition.

6.2.1.9 Permanent storage. Permanent storage is the retention of rocket motors in Magazines.

6.2.1.10 Temporary storage. Temporary storage is the retention of rocket motors in any place other than in Magazines.

6.2.1.11 Burst pressure. Burst pressure is that applied internal pressure which causes the ultimate strength of the rocket motor case to be exceeded.

6.2.1.12 Yield pressure. Yield pressure is that applied internal pressure which causes the rocket motor case to be stressed to the yield point.

6.2.1.13 Proof pressure. Proof pressure is that internal pressure to which all rocket motor cases are hydrotested.

6.2.1.14 Initial yield. Initial yield is that point at which 0.2 percent permanent deformation has occurred.

6.2.1.15 Transfer time. Transfer time is the time interval from the removal of the rocket motor or component from one conditioning chamber to the insertion of the rocket motor or component into another conditioning chamber, or the time interval from the removal of the rocket motor or component from one conditioning chamber to the initiation of testing.

6.2.1.16 Equilibrium pressure. The equilibrium pressure is the pressure level at which the rate of gas generation within the rocket motor is equal to the rate of gas flow out of the rocket motor.

6.2.1.17 Burn rate. Burn rate is the distance the propellant surface recedes in a given time interval after ignition.

6.2.1.18 Adiabatic flame temperature. The adiabatic flame temperature is the calculated temperature of the propellant combustion products within the chamber assuming no heat losses.

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6.2.1.19 Characteristic exhaust velocity. The characteristic exhaust velocity is the local sonic velocity in the exhaust nozzle throat. It is calculated by the following formula:

$$C^* = \frac{gA_t}{W_P} \int P_c dt_a .$$

6.2.1.20 Discharge coefficient. The coefficient of discharge is a correction factor and is the ratio of actual to ideal gas flow. It is calculated by the following formula:

$$C_d = \frac{F_a dt_a}{A_t C_{ftheo} \int P_c dt_a} .$$

6.2.1.21 Thrust coefficient. The thrust coefficient is an empirical constant used to relate thrust to chamber pressure for a given rocket motor design. It is calculated from measured data by the following formula:

$$C_f = \frac{\int F_a dt_a}{A_t \int P_c dt_a} .$$

6.2.1.22 Temperature coefficient of chamber pressure. The temperature coefficient of chamber pressure is the rate of change of equilibrium pressure with temperature (K_n held constant) and is calculated by the following formula:

$$k = 100 \frac{(\delta \ln Pe)}{(\delta T)} \quad K_n = \text{constant}.$$

6.2.1.23 Temperature coefficient of burn rate. The temperature coefficient of burn rate is the rate of change of burn rate with temperature (chamber pressure held constant) and is calculated by the following formula:

$$\pi P = 100 \frac{(\ln r)}{(T)} \quad P_c = \text{constant}.$$

6.2.1.24 Pressure exponent. Pressure exponent is the slope of the burn rate curve when plotted on log-log graph paper. It is calculated by the following formula:

$$n = \frac{\log r_1 - \log r_2}{\log \frac{(P_1)}{(1000)} - \log \frac{(P_2)}{(1000)}} .$$

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6.2.1.25 Area ratio. Area ratio is the quotient of the unrestricted propellant surface area at any time divided by the nozzle throat area.

6.2.1.26 Zero time. Zero time is that time at which firing voltage is applied to the igniter circuit.

6.2.1.27 Ignition delay time. The ignition delay time is the time from zero time to the point on the pressure trace when the pressure has risen to 10-percent of the maximum chamber pressure.

6.2.1.28 Ignition rise time. The ignition rise time is the time required for the pressure to rise from 10-percent of the maximum chamber pressure to 75-percent of the maximum chamber pressure.

6.2.1. 29 Action time. Action time is defined as beginning when the pressure has risen to 10-percent of the maximum chamber pressure and ending when the pressure has fallen to 10-percent of the maximum chamber pressure.

6.2.1.30 Burn time. Burn time is defined as beginning when the pressure has risen to 10-percent of the maximum chamber pressure and ending when the pressure has dropped to 75-percent of the maximum chamber pressure.

6.2.1.31 Maximum chamber pressure. The maximum chamber pressure is the highest chamber pressure developed by the rocket motor under any normal operating condition excluding ignition.

6.2.1.32 Maximum ignition pressure. The maximum ignition pressure is the highest chamber pressure developed by the rocket motor during ignition.

6.2.1.33 Average chamber pressure. The average chamber pressure is the area under the pressure-time curve between the 10-percent points defined in 6.2.1.29 divided by the action time.

6.2.1.34 Maximum thrust. The maximum thrust is the highest thrust developed by the rocket motor under any normal operating condition excluding ignition.

6.2.1.35 Maximum ignition thrust. The maximum ignition thrust is the highest thrust developed by the rocket motor during ignition.

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6.2.1.36 Average thrust. The average thrust is the action time total impulse divided by the action time.

6.2.1.37 Action time total impulse. The action time total impulse is the area under the thrust-time curve between the 10-percent points defined in 6.2.1.29.

6.2.1.38 Burn time total impulse. The burn time total impulse is the area under the thrust-time curve between the points defined in 6.2.1.30.

6.2.1.39 Deliverable total impulse. The deliverable total impulse is the total area under the thrust-time curve.

6.2.1.40 Overall specific impulse. Overall specific impulse equals the deliverable total impulse divided by the rocket motor weight.

6.2.1.41 Propellant specific impulse. Propellant specific impulse equals the deliverable total impulse divided by the total propellant weight.

6.2.1.42 List of ballistic parameter symbols. See pages 24, 25 & 26.

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BALLISTIC PARAMETERS SYMBOLS LIST

Symbol	Quantity	Unit
A	Area	in^2
A_e	Exhaust Nozzle Exit Area	in^2
A_p	Port Area	in^2
A_t	Exhaust Nozzle Throat Area	in^2
C_d	Discharge Coefficient	--------
C_f	Thrust Coefficient	--------
C^*	Characteristic Exhaust Velocity	ft/sec
F_a	Thrust	lb
$\overline{F}_a$	Average Thrust	lb
F_i	Maximum Ignition Thrust	lb
F_{max}	Maximum Thrust	lb
F_{sl}	Sea Level Thrust	lb
g	Gravitational Constant	ft/sec^2
I_a	Action Time Total Impulse	lb-sec
I_{alt}	Impulse at Altitude	lb-sec
I_b	Burn Time Total Impulse	lb-sec
I_d	Deliverable Total Impulse	lb-sec
I_{oa}	Overall Specific Impulse	$\frac{\text{lb-sec}}{\text{lb}}$
I_{sl}	Impulse at Sea Level	lb-sec
I_{sp}	Propellant Specific Impulse	$\frac{\text{lb-sec}}{\text{lb}}$

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BALLISTIC PARAMETERS SYMBOLS LIST - Contd -

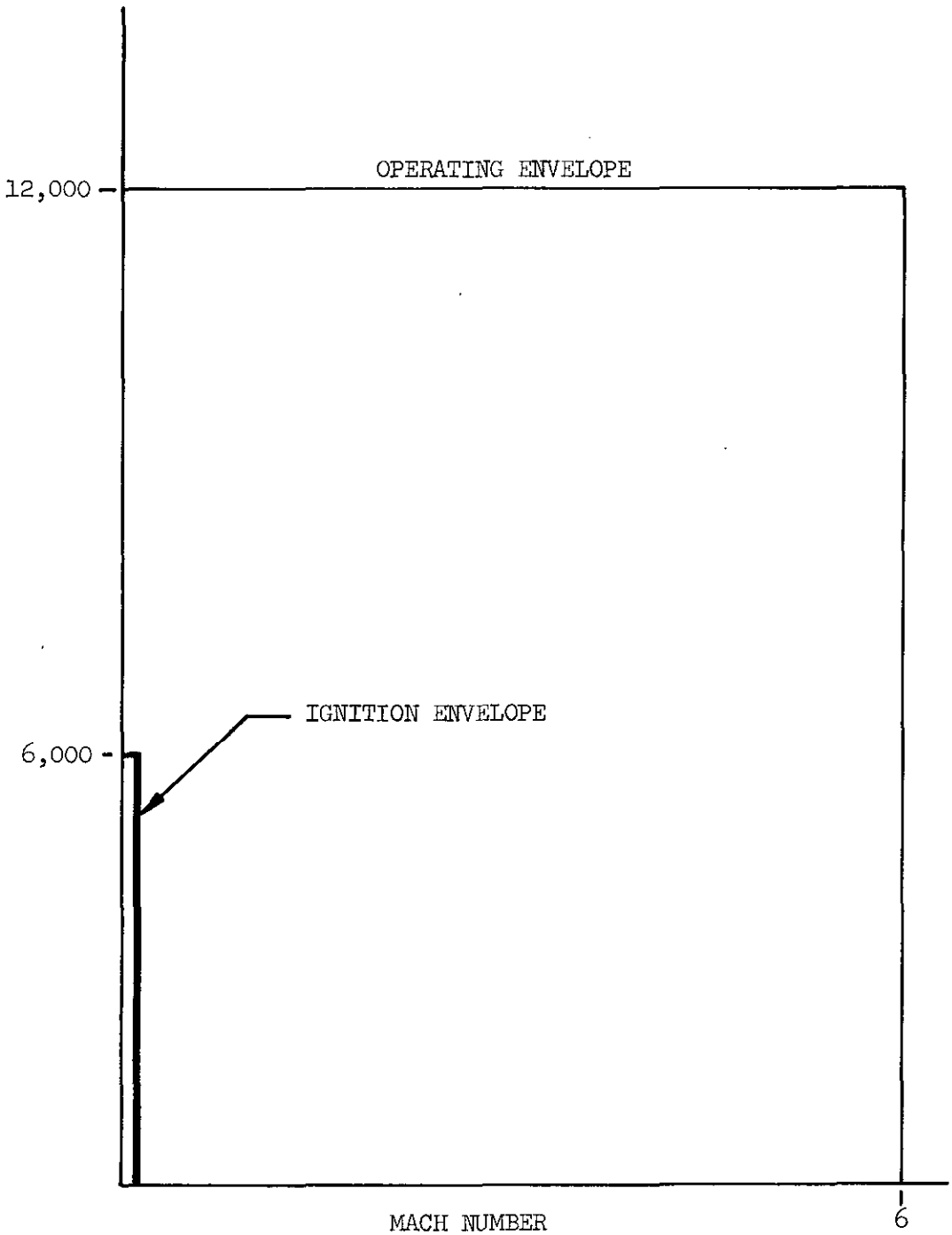
Symbol	Quantity	Unit
K	Area Ratio	--------
ln	Logarithm to the Base e	--------
log	Logarithm to the Base 10	--------
n	Pressure Exponent	--------
n_x	Fore and Aft Load Factor	--------
n_y	Side Load Factor	--------
$\overline{P}_a$	Average Chamber Pressure	psia
P_{alt}	Altitude Barometric Pressure	psia
P_c	Chamber Pressure	psia
P_e	Equilibrium Pressure	psia
P_i	Maximum Ignition Pressure	psia
P_{max}	Maximum Chamber Pressure	psia
P_{sl}	Sea Level Barometric Pressure	psia
r	Burn Rate	in/sec
t_a	Action Time	sec
t_b	Burn Time	sec
t_d	Ignition Delay Time	sec
t_o	Zero Time	--------
t_r	Ignition Rise Time	sec
T	Temperature	°F
W_m	Total Weight of Rocket Motor	lb

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BALLISTIC PARAMETERS SYMBOLS LIST - Cont -

Symbol	Quantity	Unit
W_p	Propellant Weight	lb
k	Temperature Coefficient of Chamber Pressure	$\frac{\%}{°F}$
p	Temperature Coefficient of Burn Rate	$\frac{\%}{°F}$
'ɵ'	Pitching Acceleration	rad/sec^2
' '	Yawing Acceleration	rad/sec^2

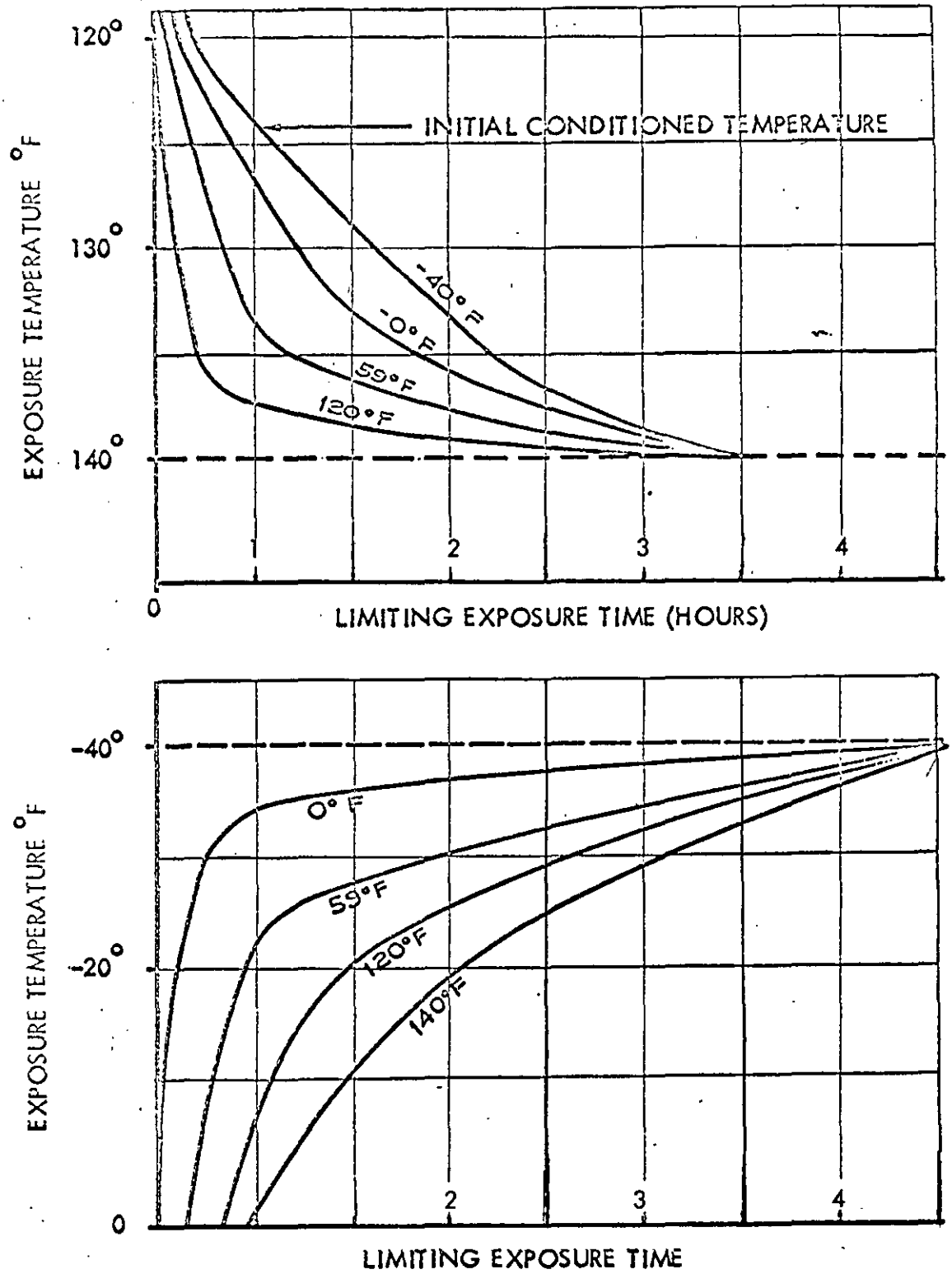
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IGNITION AND OPERATING ENVELOPES

FIGURE 1

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LIMITING EXPOSURE TIME AFTER EXTREME TEMPERATURE EXPOSURE

FIGURE 2

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SEA LEVEL CHAMBER PRESSURE AND

THRUST VS. TIME AT -40°F.

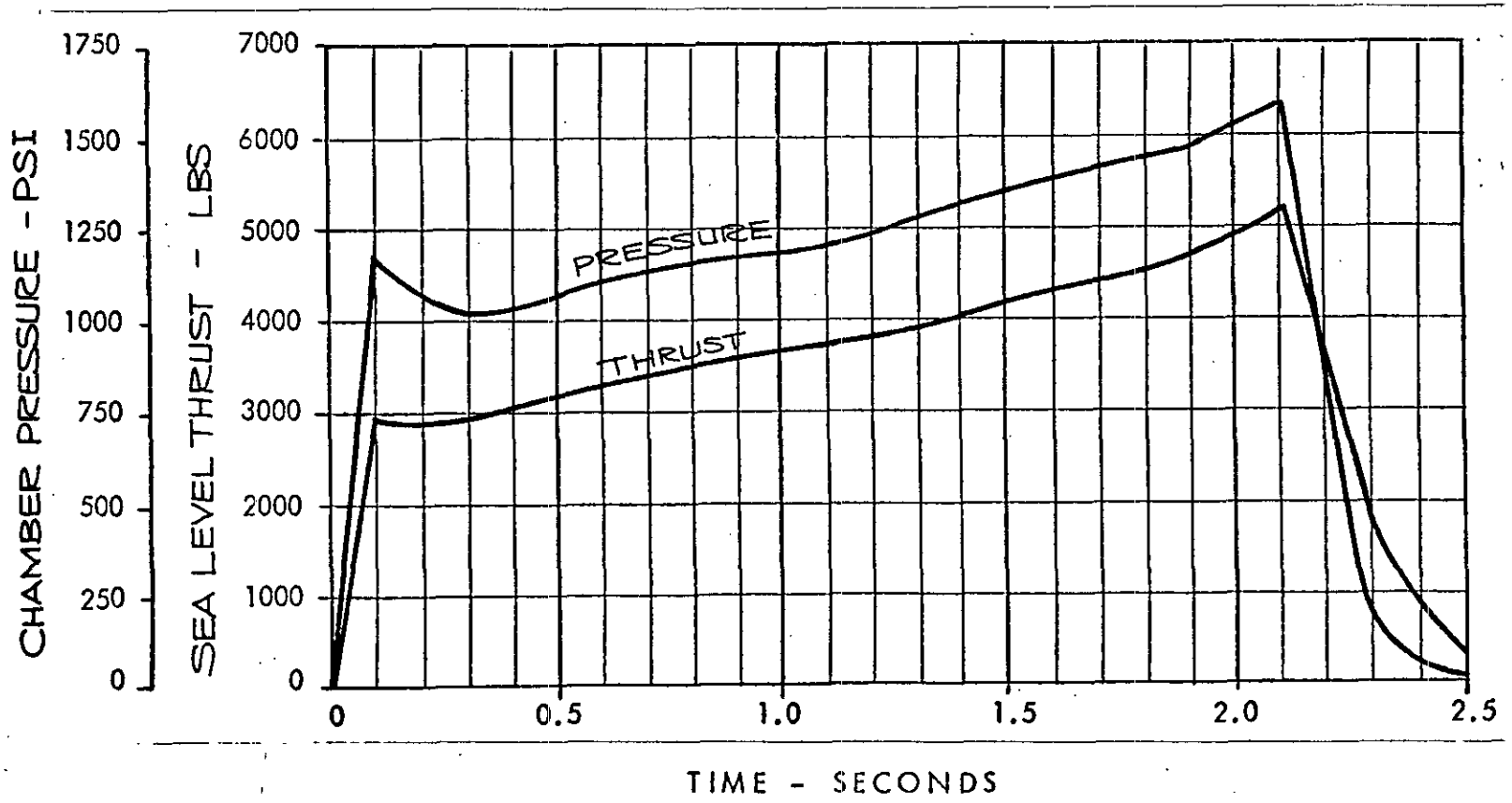


FIGURE 3

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SEA LEVEL CHAMBER PRESSURE AND THRUST VS. TIME AT +59°F.

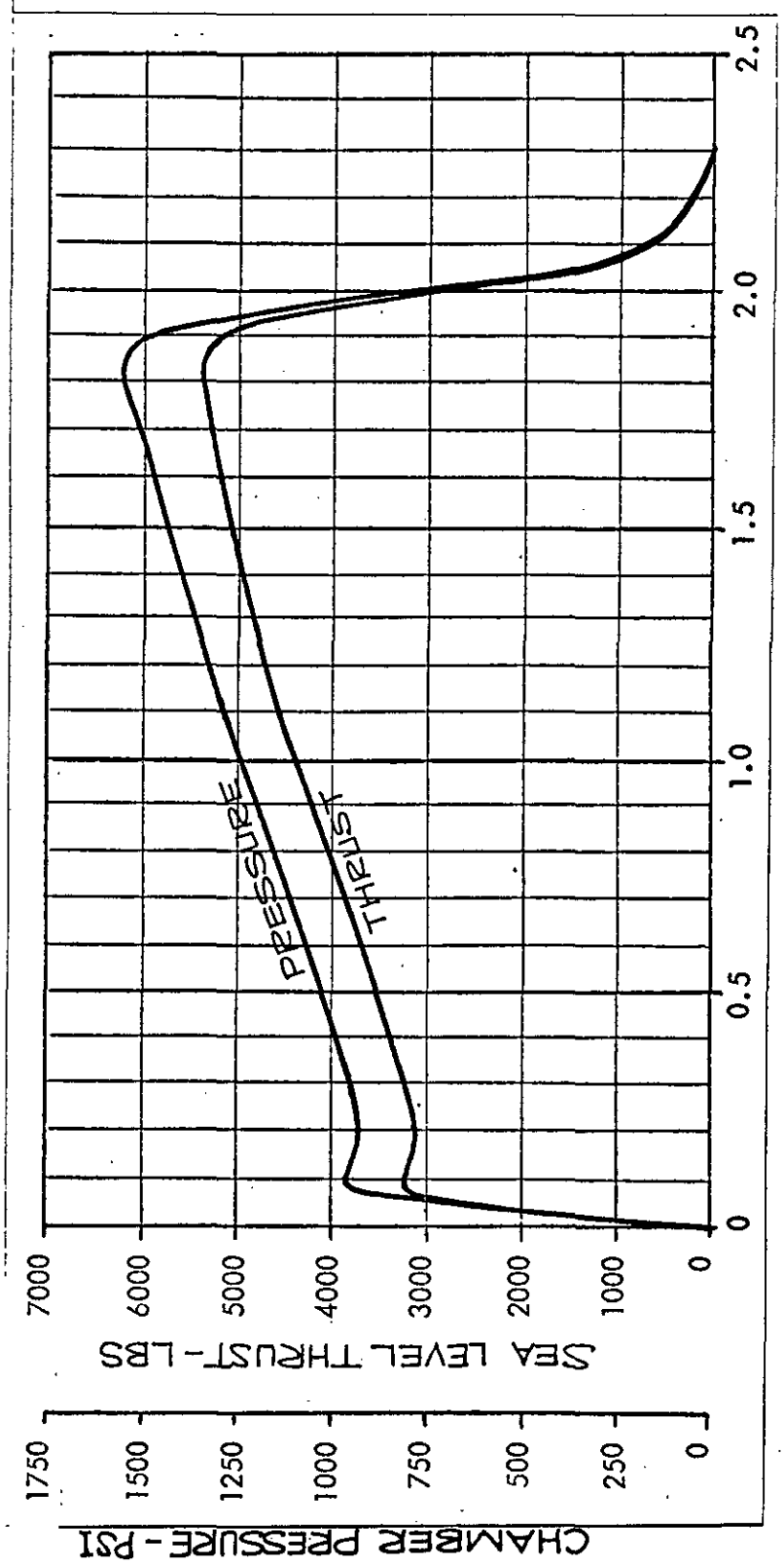
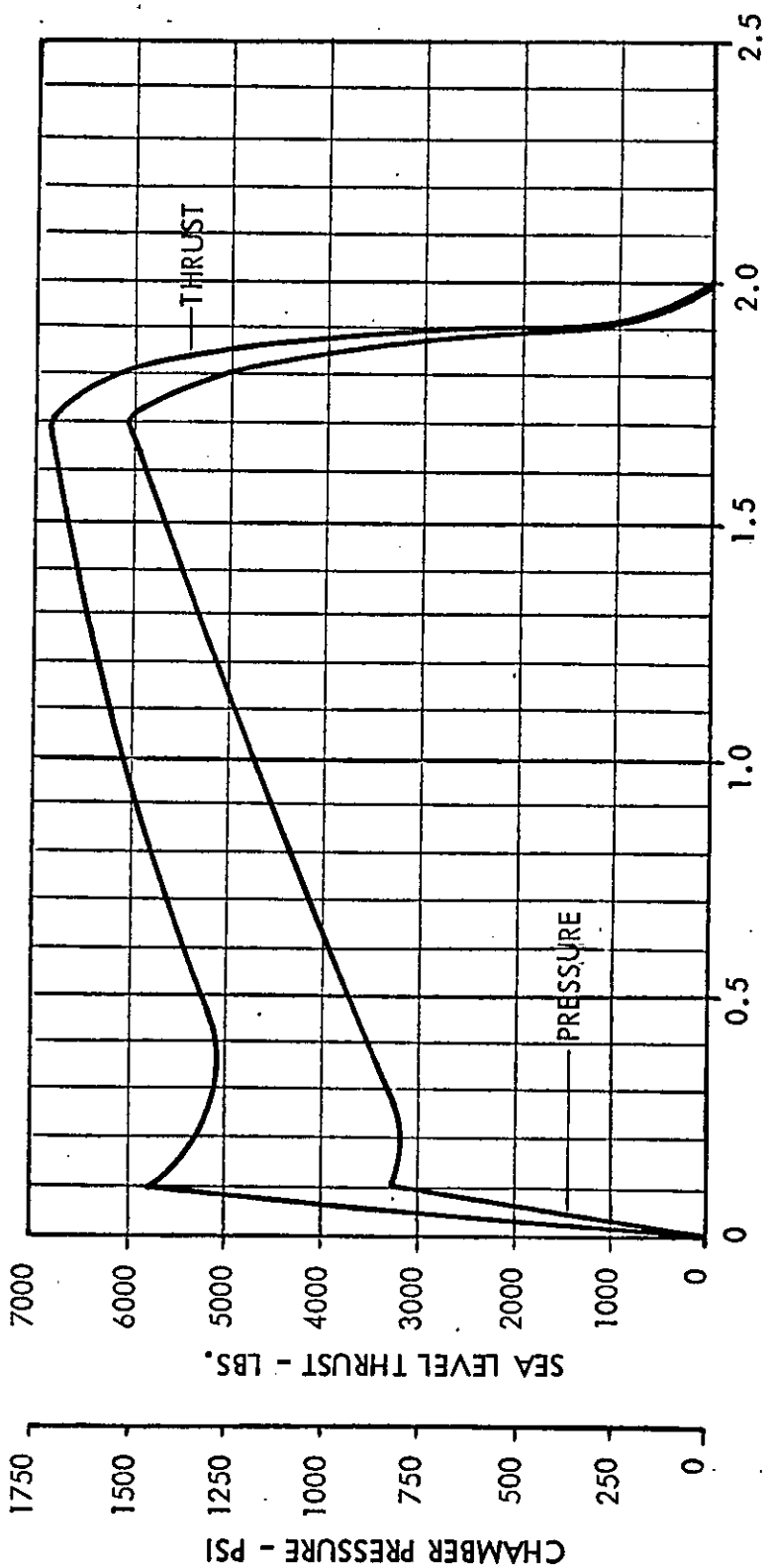


FIGURE 4

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FIGURE 5

SEA LEVEL CHAMBER PRESSURE AND THRUST VS. TIME AT +140° F.



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VARIATION OF AVERAGE THRUST WITH ROCKET MOTOR TEMPERATURE AT SEA LEVEL

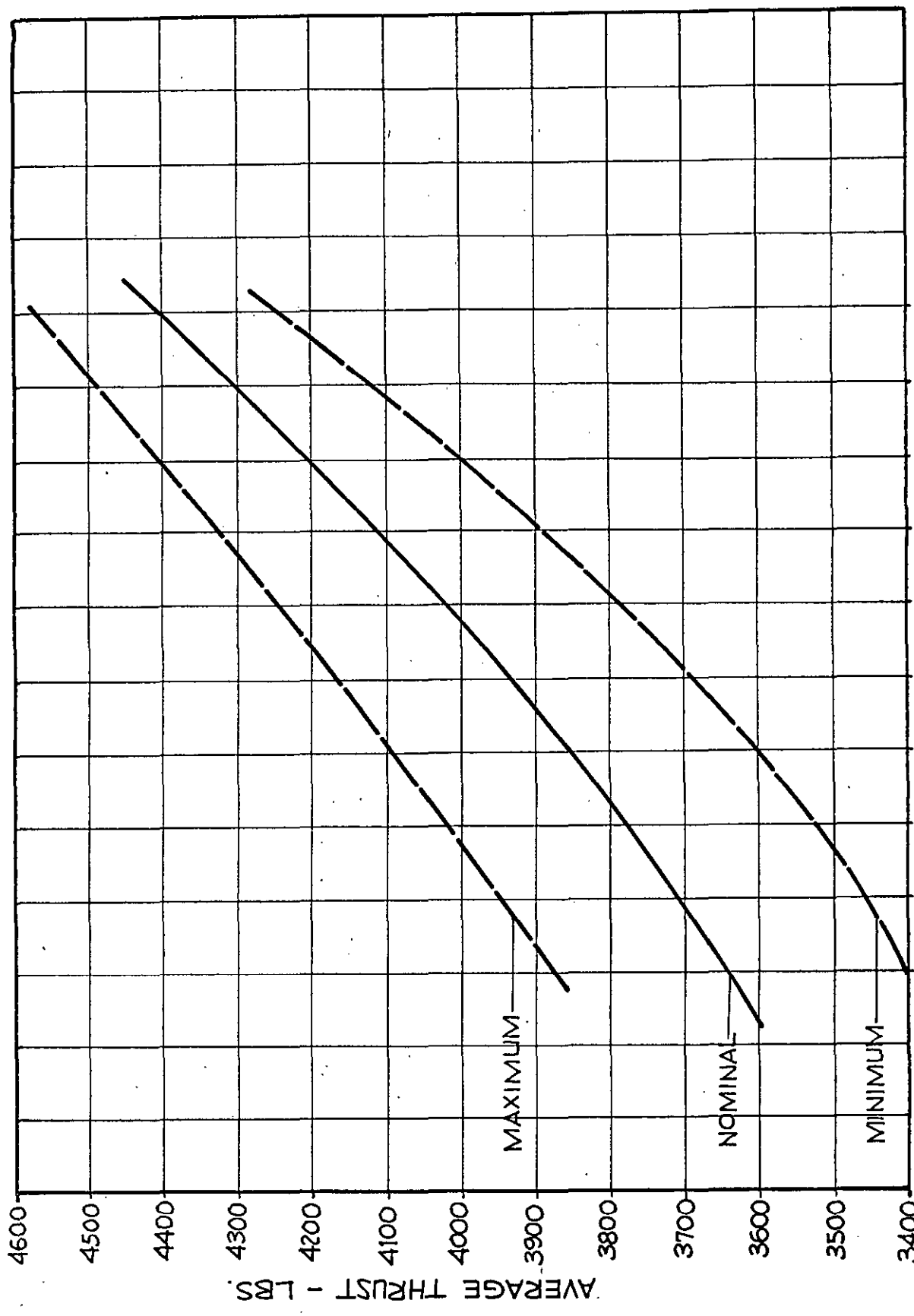
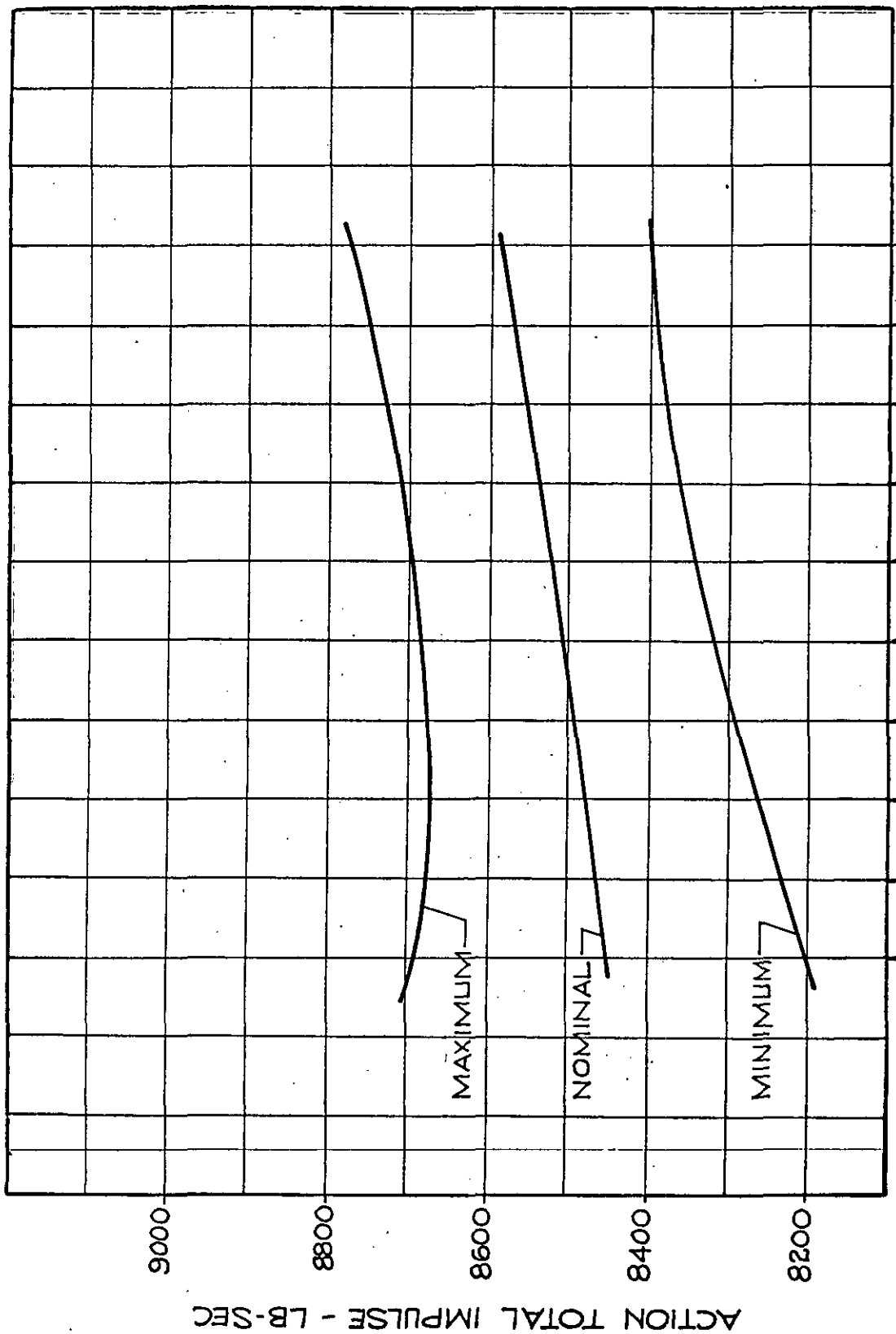


FIGURE 6

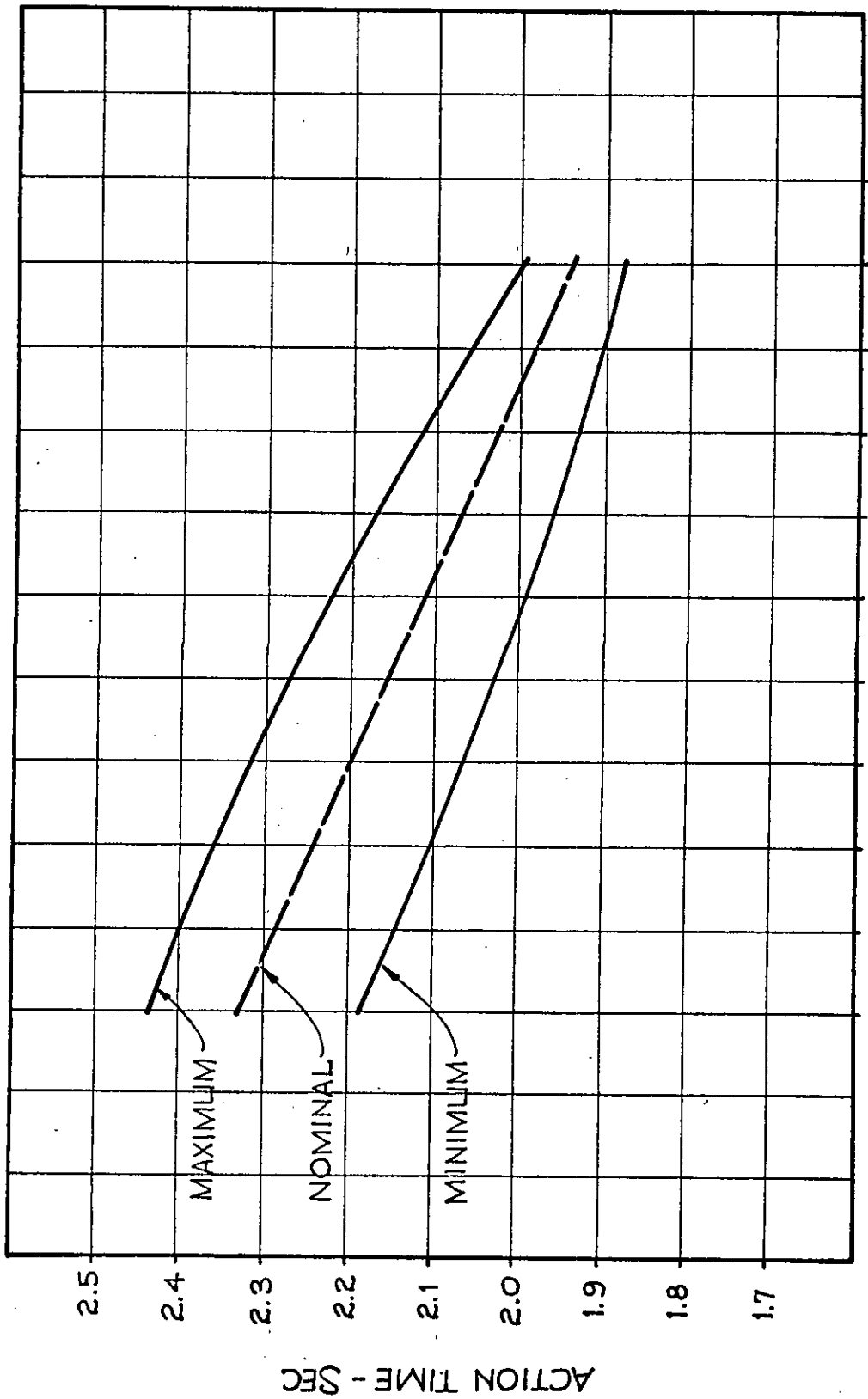
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VARIATION OF ACTION TIME TOTAL IMPULSE WITH ROCKET MOTOR TEMPERATURE AT SEA LEVEL

FIGURE 7

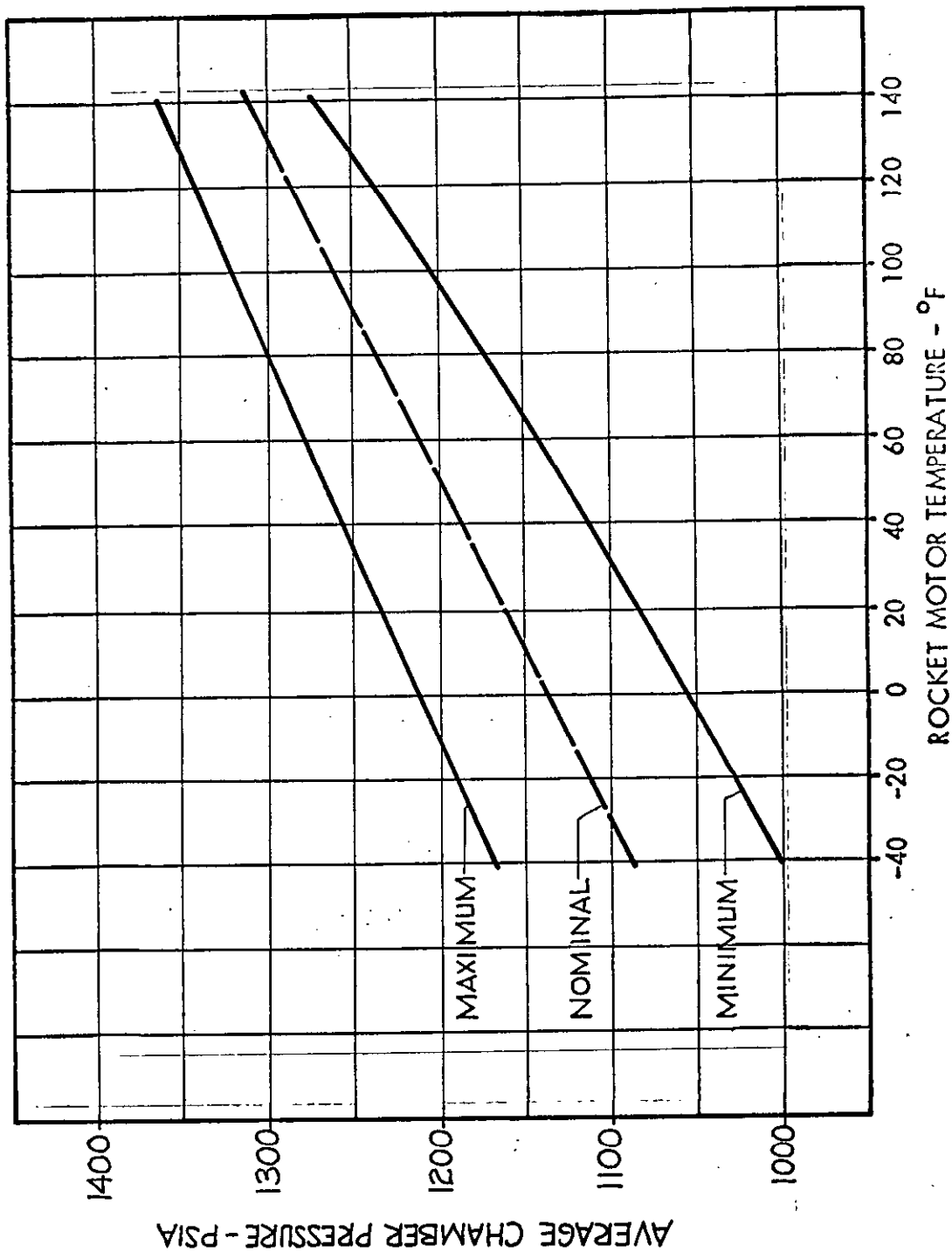
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VARIATION OF ACTION TIME WITH ROCKET MOTOR TEMPERATURE AT SEA LEVEL

FIGURE 8

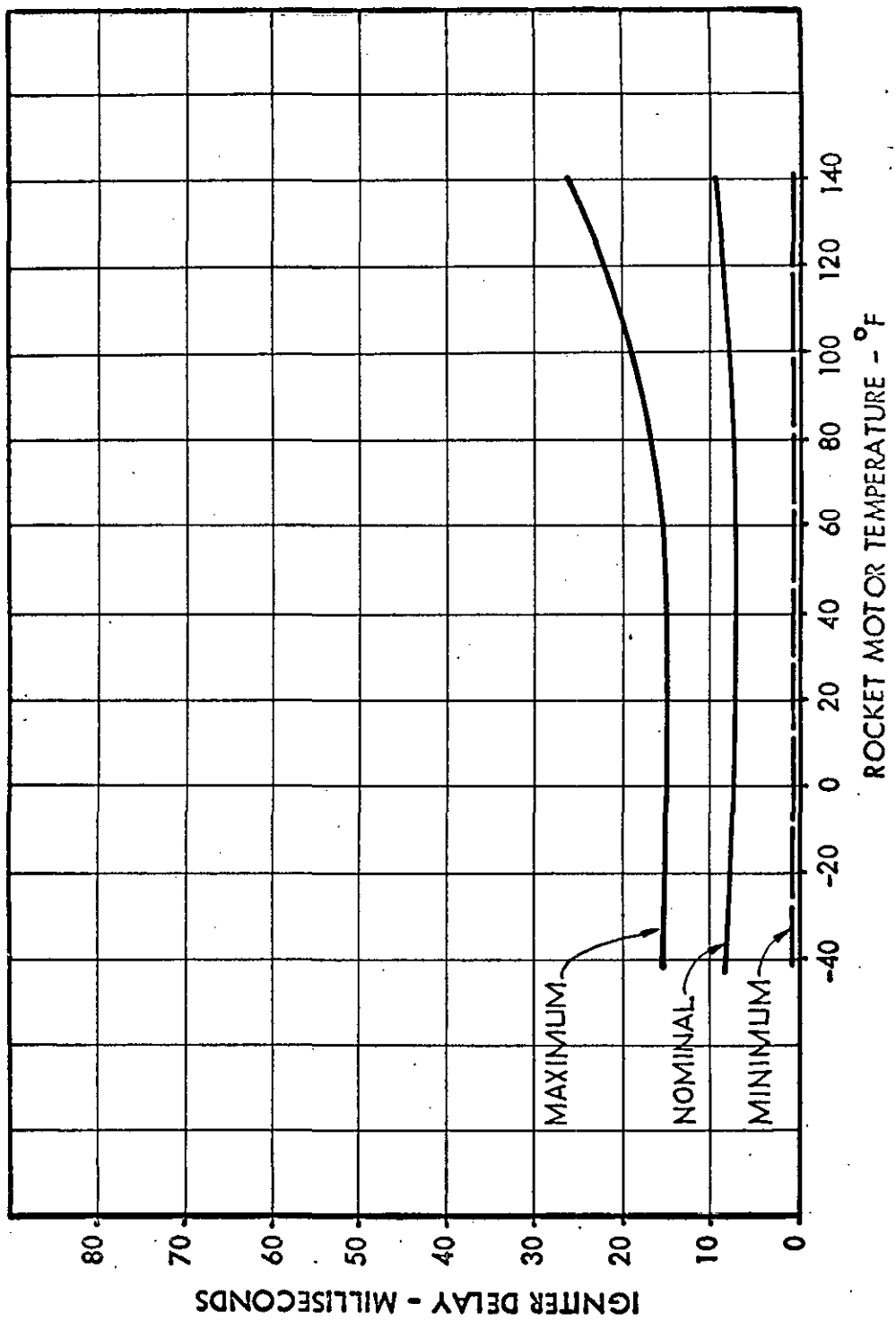
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VARIATION OF AVERAGE CHAMBER PRESSURE WITH ROCKET MOTOR TEMPERATURE AT SEA LEVEL

FIGURE 9

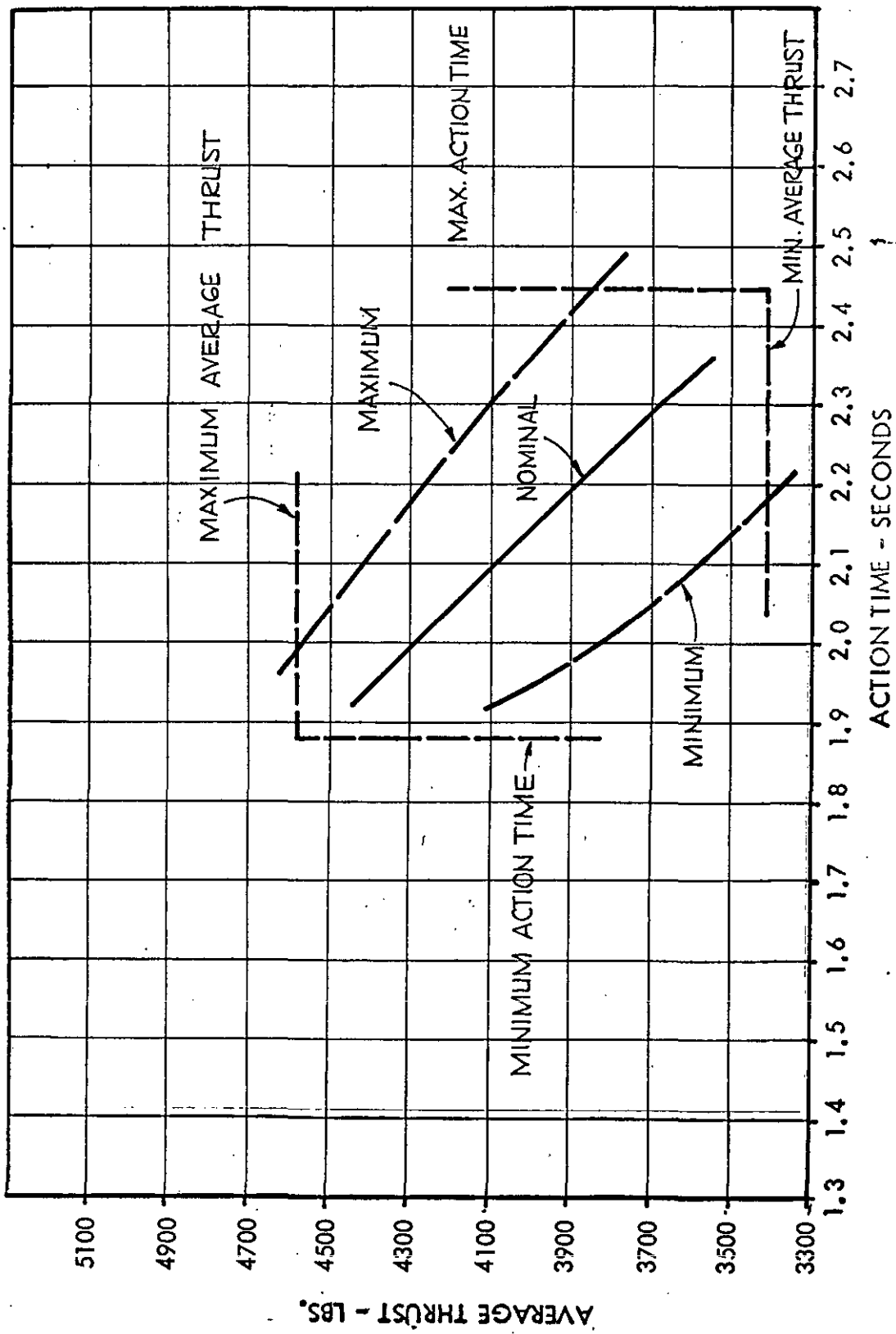
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VARIATION OF IGNITION DELAY WITH ROCKET MOTOR TEMPERATURE AT SEA LEVEL

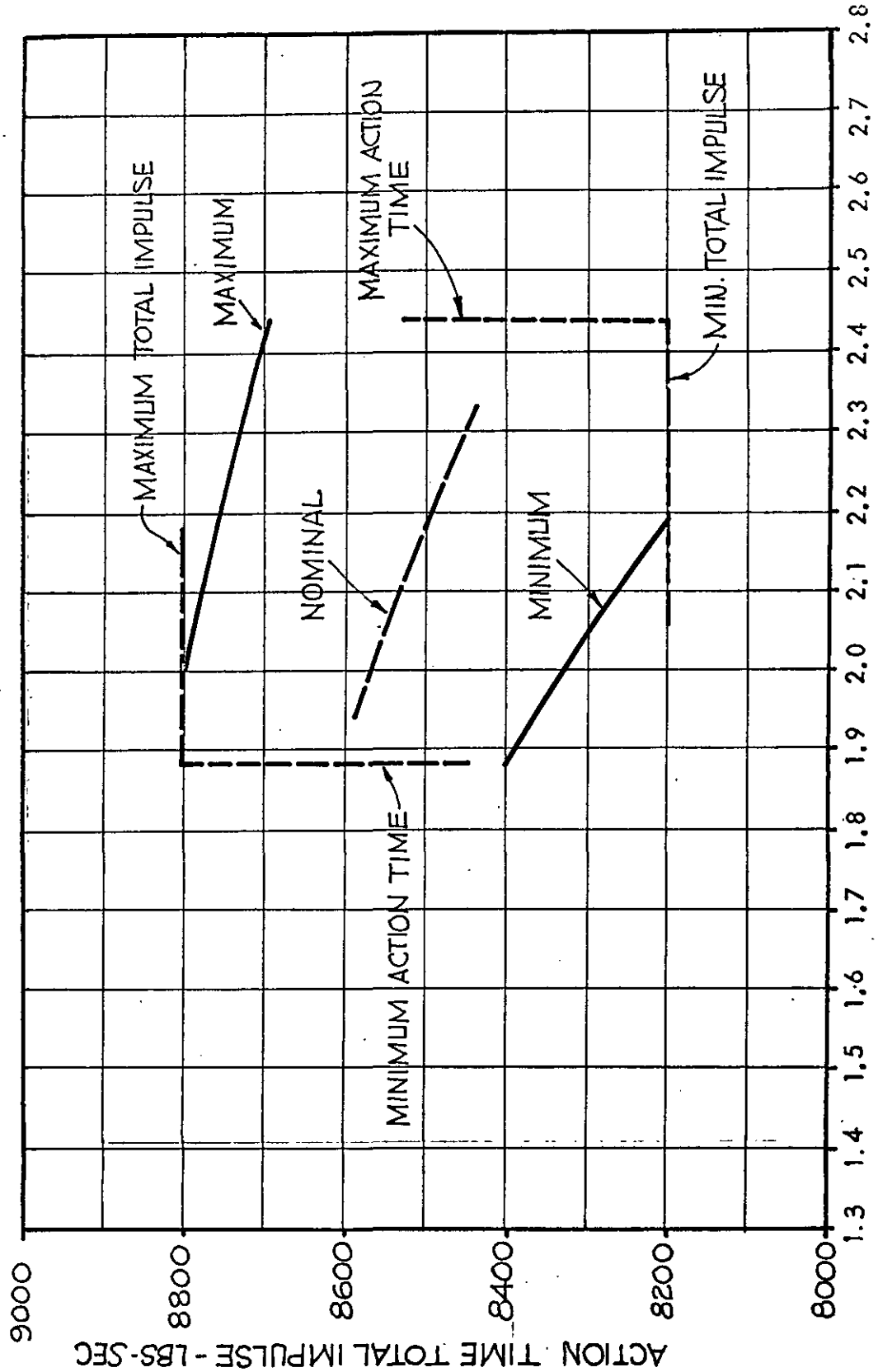
FIGURE 10

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VARIATION OF AVERAGE THRUST WITH ACTION TIME
AT SEA LEVEL
FIGURE 11

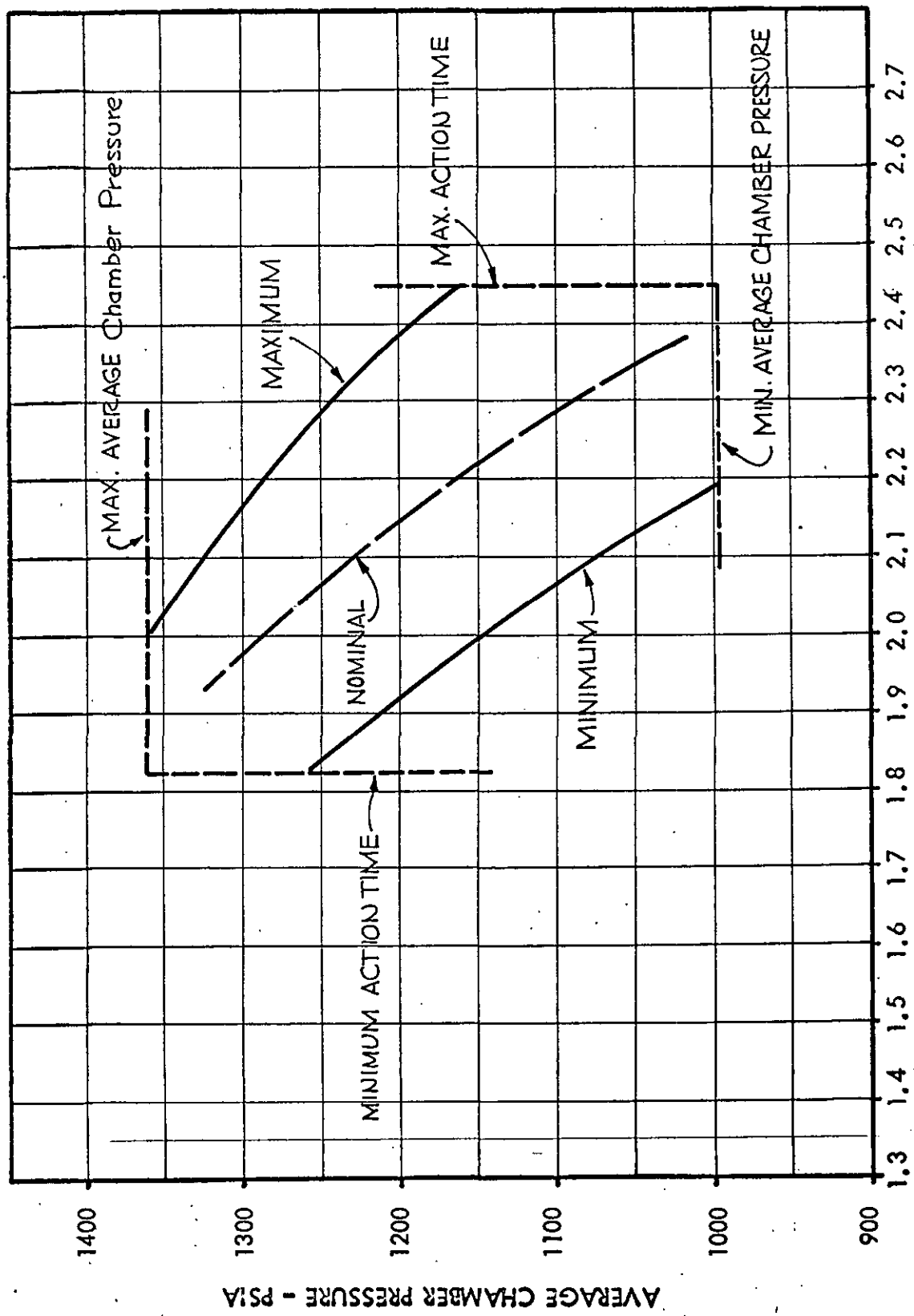
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VARIATION OF ACTION TIME TOTAL IMPULSE WITH ACTION TIME AT SEA LEVEL

FIGURE 12

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VARIATION OF AVERAGE CHAMBER PRESSURE WITH ACTION TIME AT SEA LEVEL

FIGURE 13

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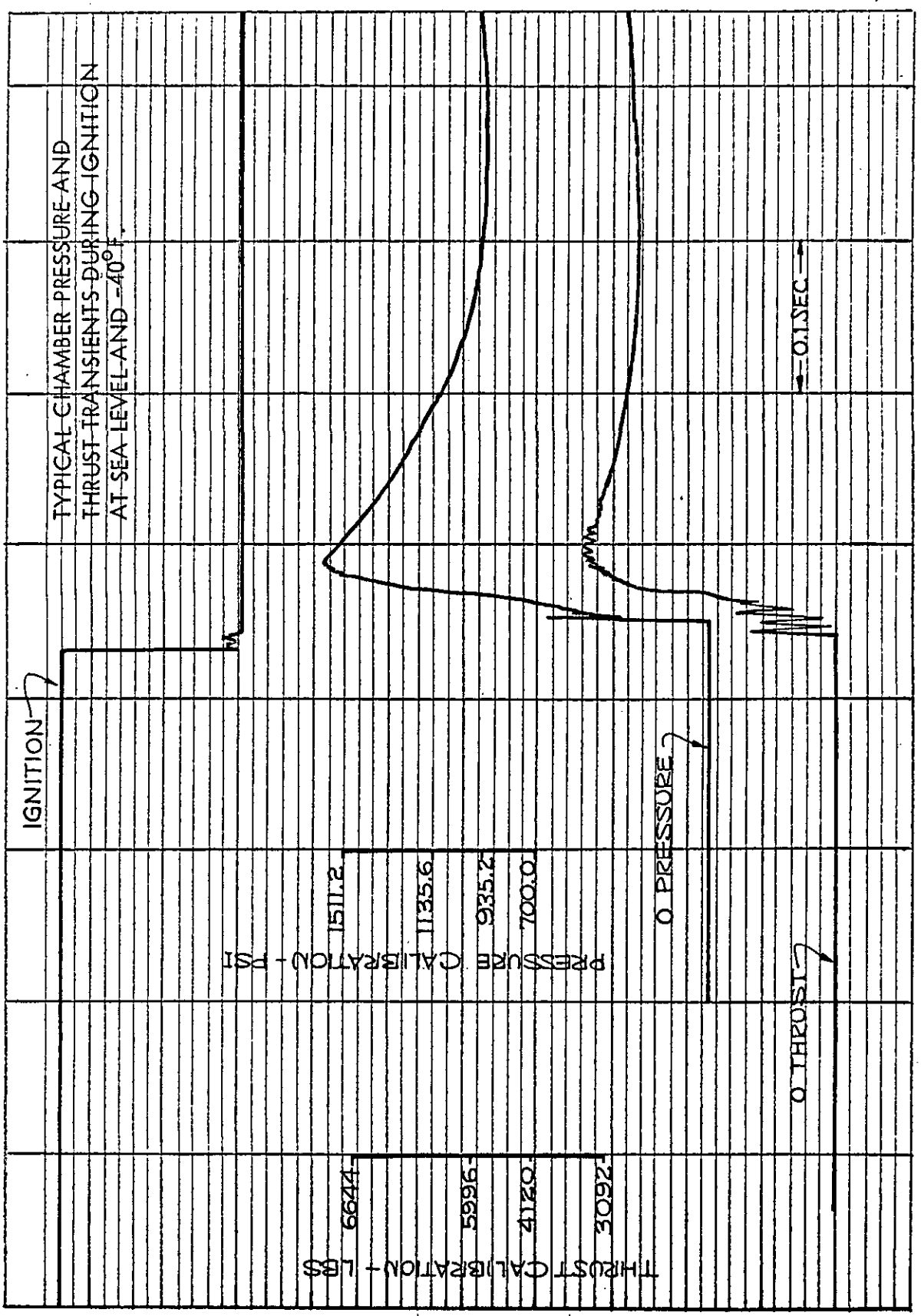


FIGURE 14

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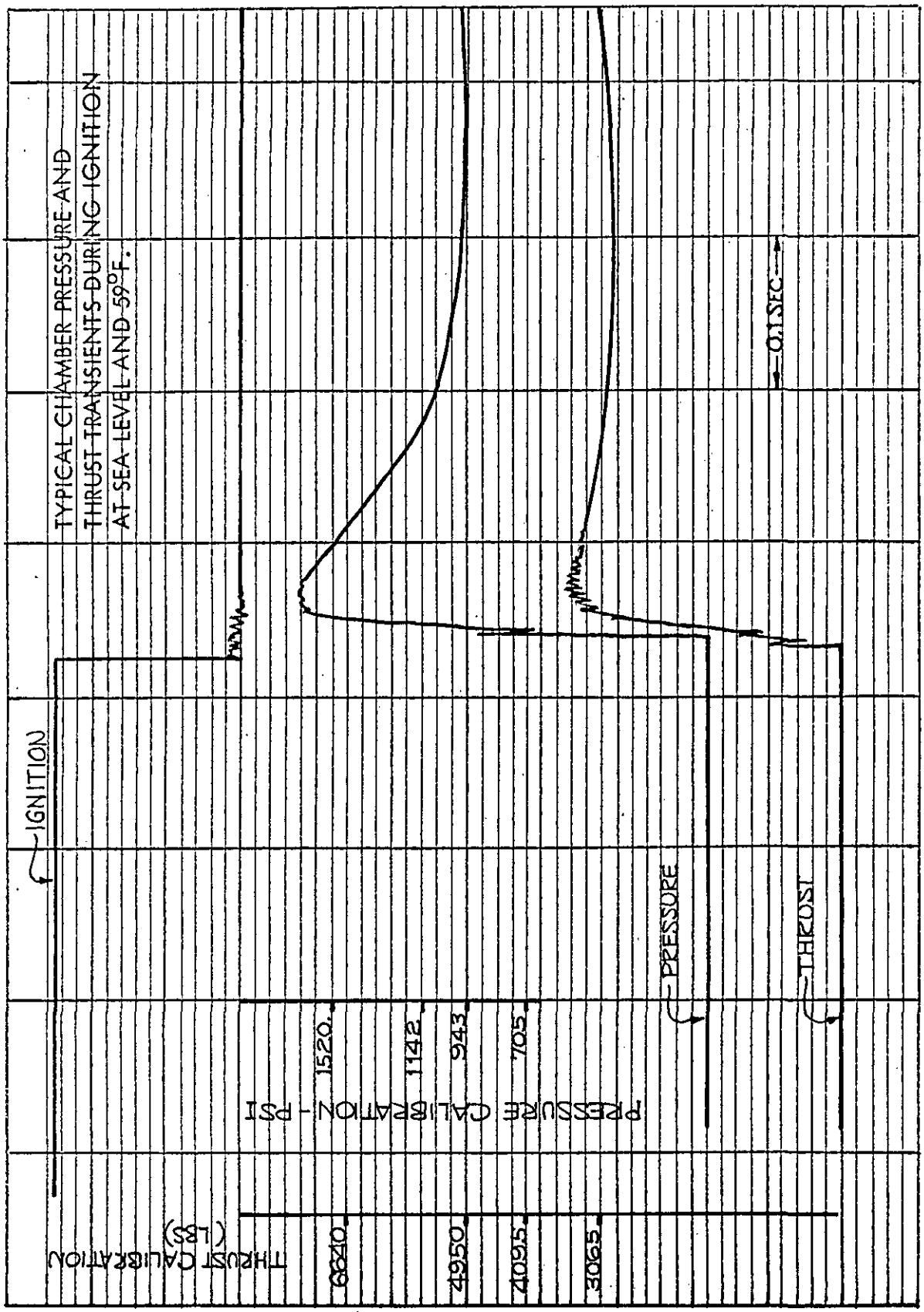


FIGURE 15

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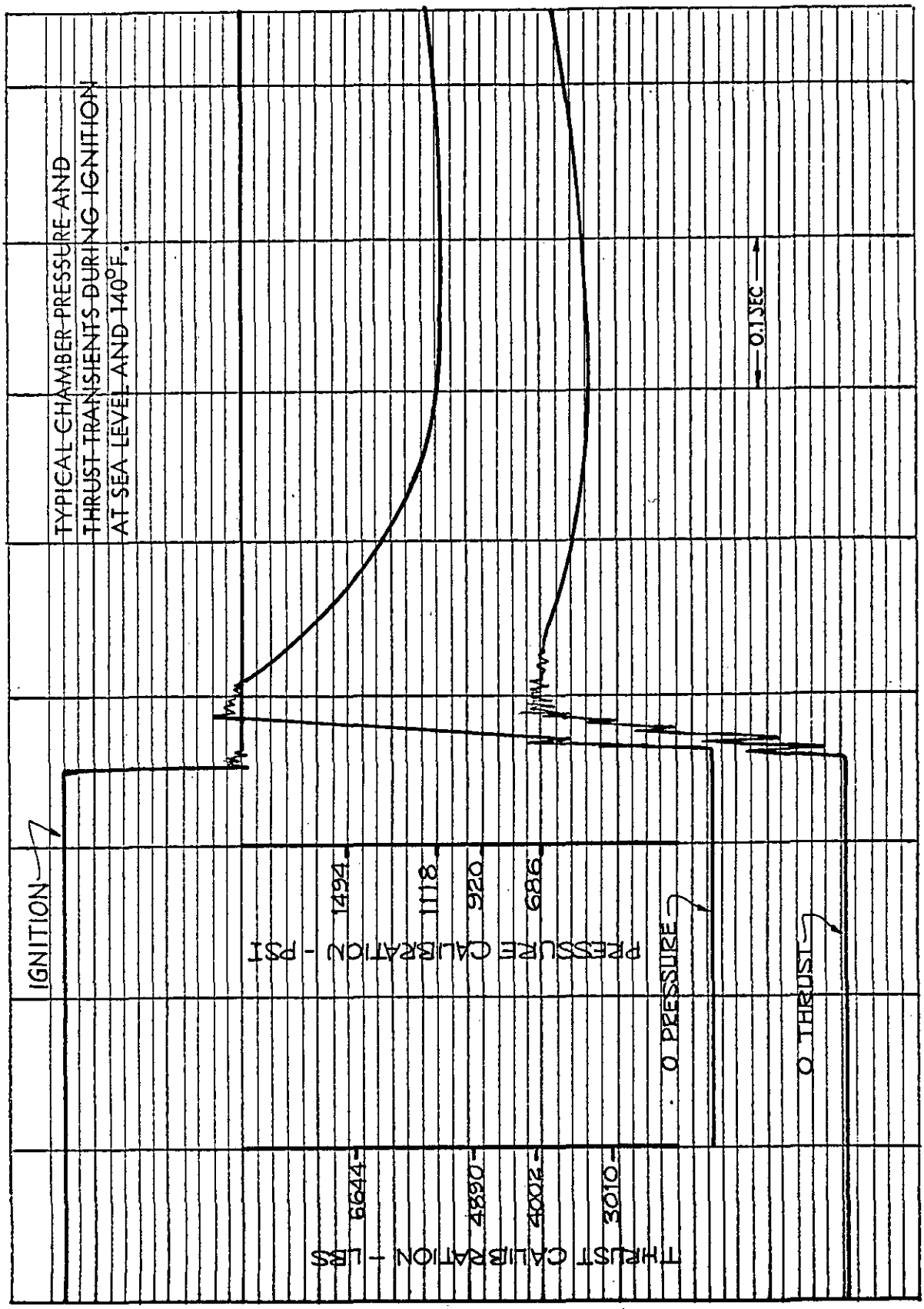
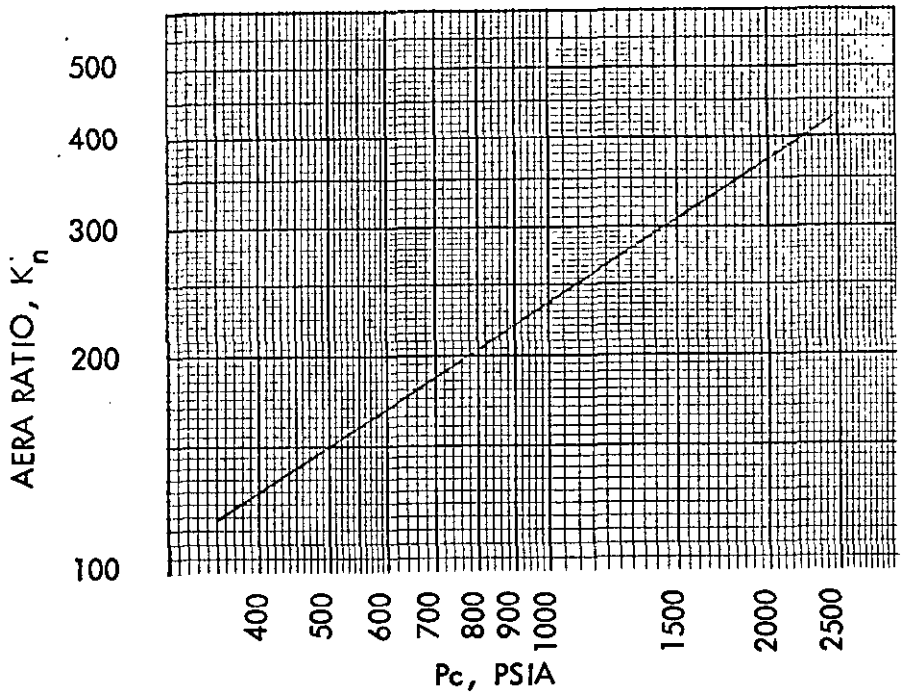


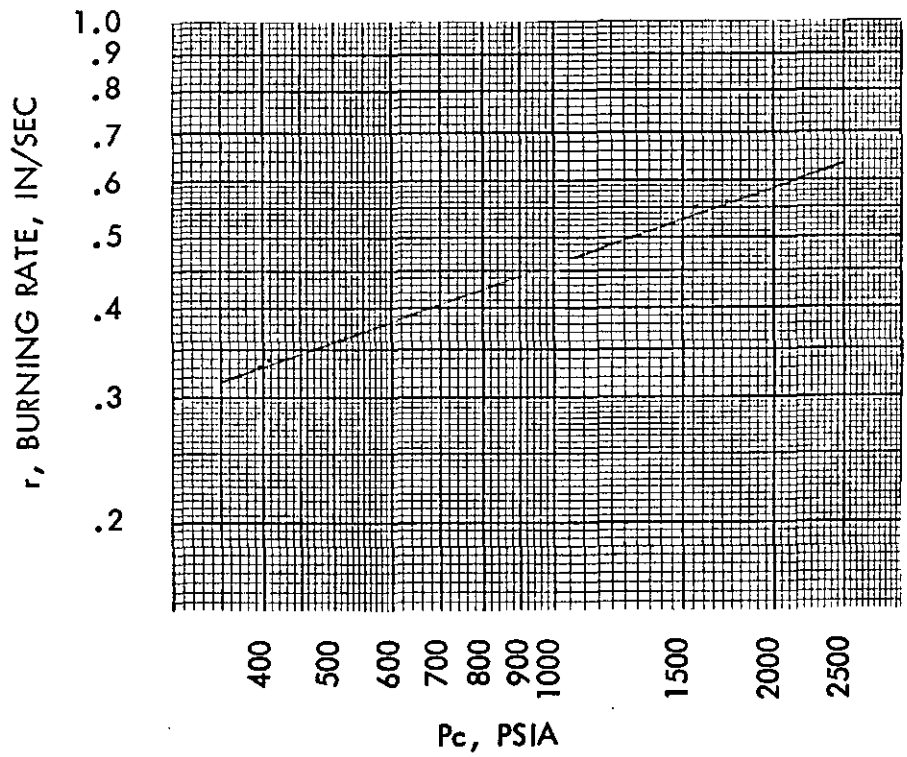
FIGURE 16

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$K_n = 2.675P_c^{0.648}$



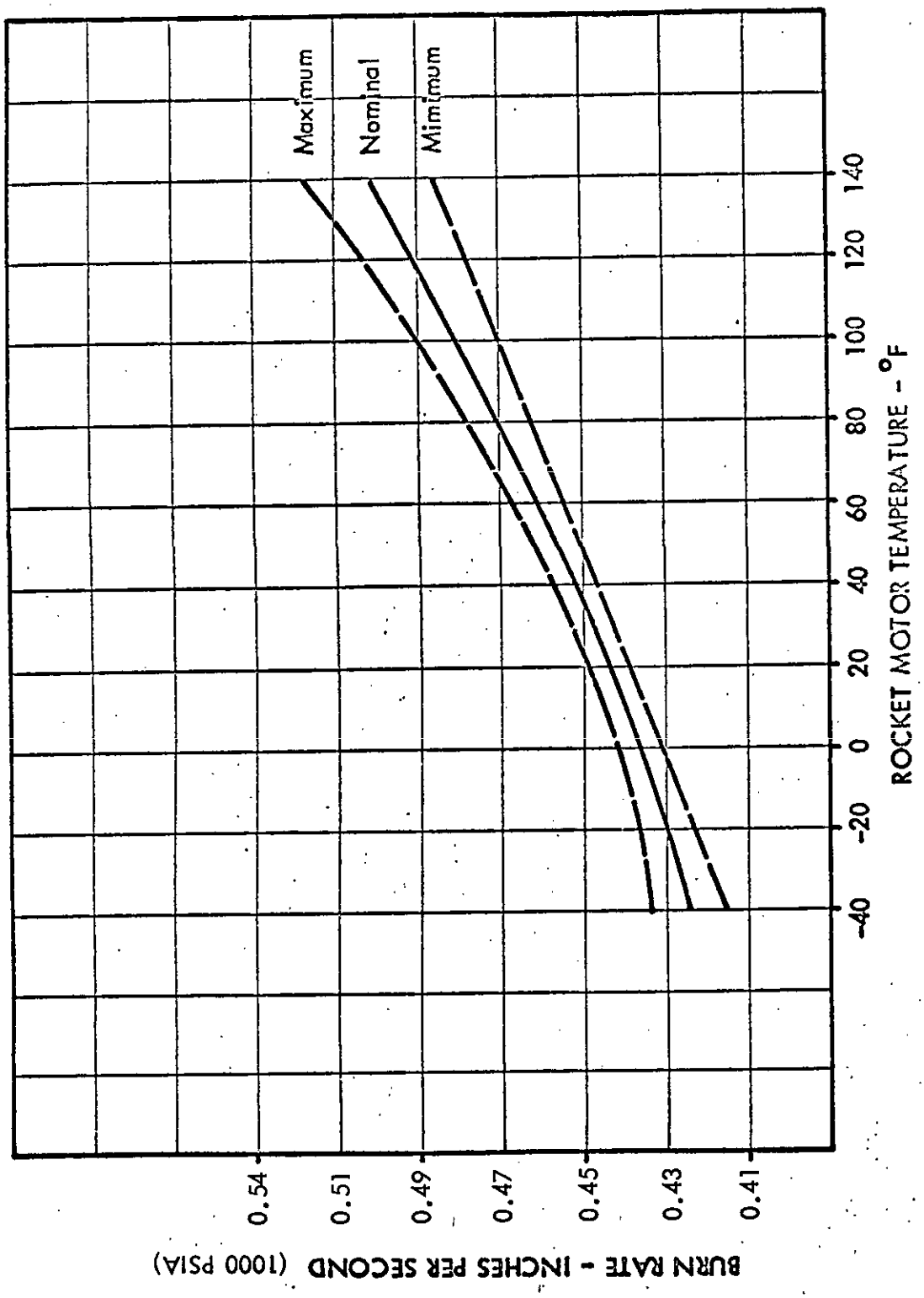
$r = .0404P_c^{0.352}$



BURNING RATE DATA AMBIENT TEMPERATURES

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FIGURE 17



VARIATION OF BURN RATE WITH ROCKET MOTOR TEMPERATURE

FIGURE 18

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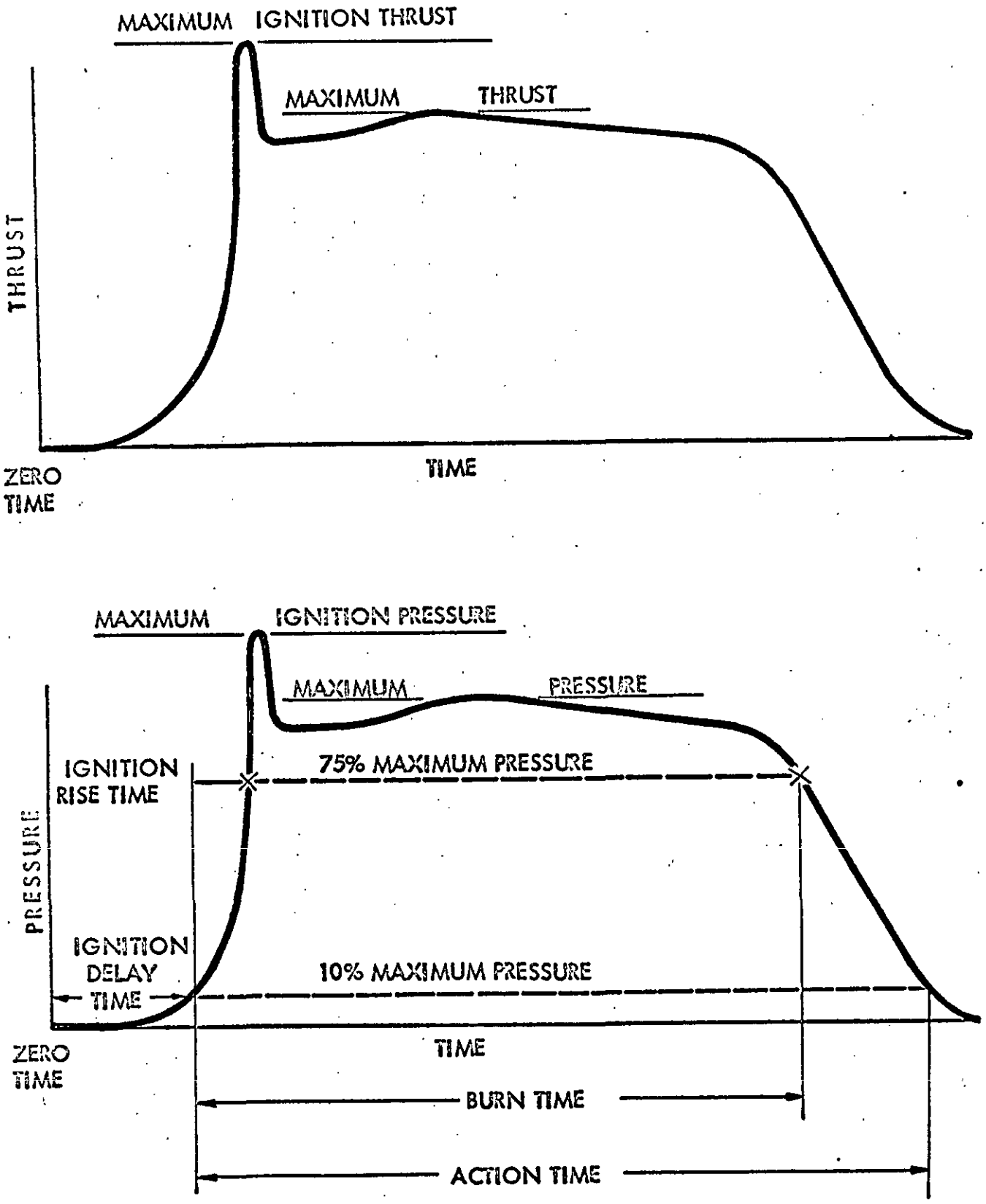


ILLUSTRATION OF BALLISTIC DEFINITIONS

FIGURE 19

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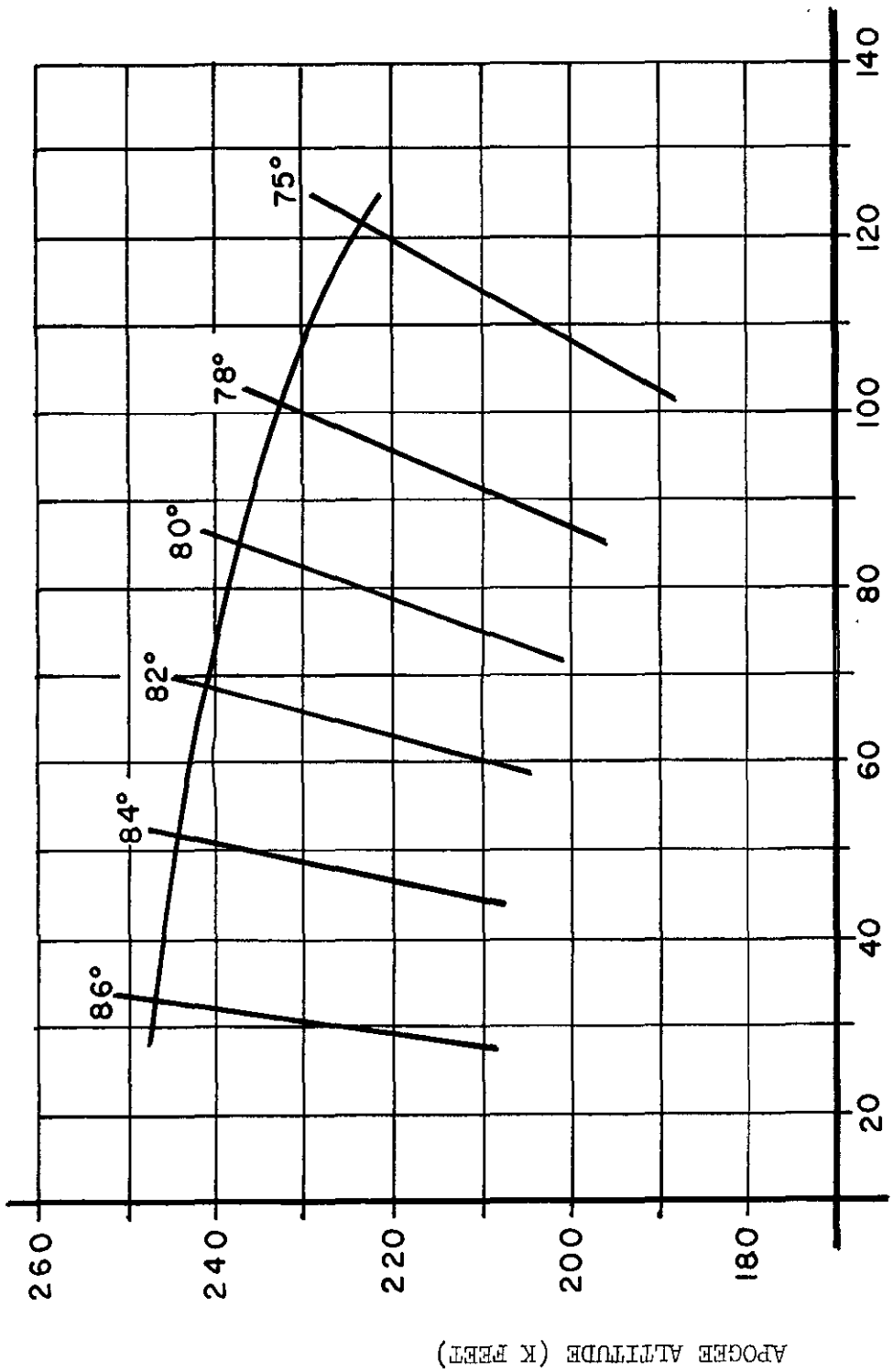


FIGURE 20. PERFORMANCE SPECIFICATIONS
METEOROLOGICAL PROBE PWN-10A, -10B, -11A

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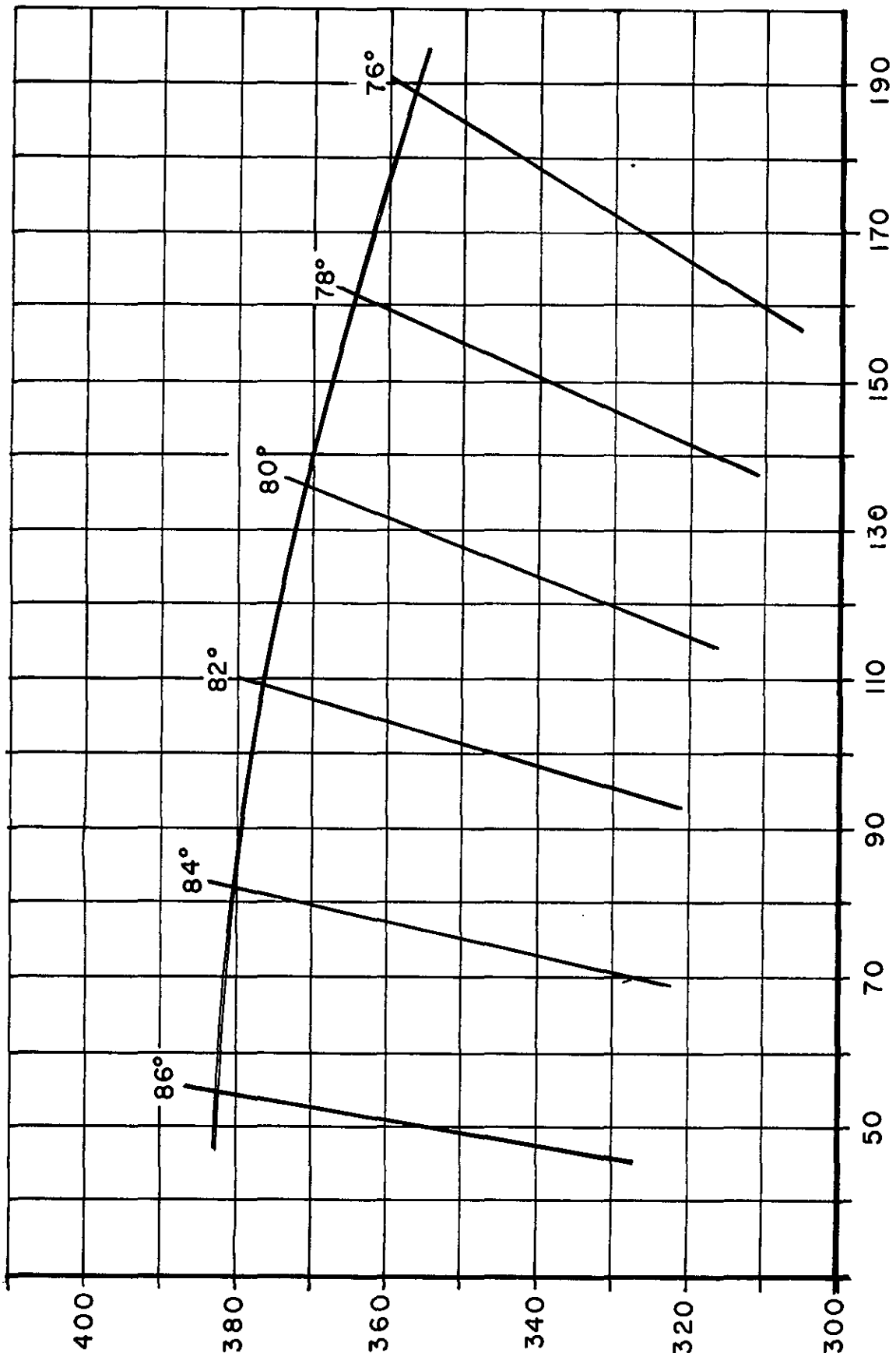


FIGURE 21. PERFORMANCE SPECIFICATIONS
METEOROLOGICAL PROBE PWN-12A

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Custodian:

Air Force - 13

Review Activity:

Air Force - 70

Preparing Activity:

Air Force - 13

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